Do Tail-Risk Penalties Help Deep RL Portfolio Allocators? A Walk-Forward Evaluation with Tradable Execution, Market Impact and Pre-Specified Inference

Deborshi Vikramaditya Chakravorty

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Abstract

Deep reinforcement learning (DRL) portfolio allocators are often given tail-risk penalties in their reward to make them safer. We evaluate that practice for proximal policy optimisation (PPO) on ten liquid multi-asset ETFs over 14 annual walk-forward folds (October 2012 to October 2026, 3,527 test days). The simulator charges commission, half-spread and square-root market impact; every design is trained with five seeds; and every strategy, including classical rules, an untrained-policy control and an exposure-matched control, runs through the same simulator. Orders are decided after the close and filled at the next open. An audit found that earlier versions of this study filled orders at the same close that the features had already used, which cannot be traded. We corrected the simulator, reran the decisive experiments, and tested five comparisons fixed in an analysis plan committed before the rerun, using a two-level (seed and time-block) bootstrap with Holm correction.

Under tradable execution, vanilla PPO earns a Sharpe ratio of 0.11 against 0.64 for equal weighting (difference −0.53, 95% CI [−0.79, −0.32]). Calibrated CVaR and drawdown penalties lower daily CVaR and turnover but have no detectable effect on the Sharpe ratio (+0.06 [−0.12, 0.26]). Partial-adjustment execution is the only design choice with a significant effect (+0.49 [0.27, 0.74]); the best agent is then neither distinguishable from nor equivalent to 1/N (−0.08 [−0.23, 0.06]). The pre-cost timing edge reported in earlier versions does not survive (−0.09 [−0.30, 0.12]): it came from the non-tradable fill. No agent de-risks when volatility rises, 1/N delevered to each agent's cash level matches its drawdown, and the untrained policy, which holds equal weights in every asset and cash, is as good as any trained agent. No agent beats 1/N even with every cost removed, and the conclusions hold under other impact and fill assumptions, with from-scratch training, and in four independent simulated markets, one of them a bear market. The framework, tests, analysis plan and all outputs are public.

Keywords: deep reinforcement learning; proximal policy optimization; conditional value-at-risk; drawdown; market impact; execution timing; look-ahead bias; walk-forward validation; multiple testing; portfolio allocation

1 Introduction

Classical portfolio construction solves a static problem. Mean-variance optimisation [1] and its descendants choose weights from estimated moments and rebalance periodically. Estimation error in expected returns is large enough that the naive 1/N rule is hard to beat out of sample [4], so practitioners lean on shrinkage [5] and on risk-based allocations such as equal risk contribution [6] and hierarchical risk parity [8].

Deep reinforcement learning offers a state-dependent, non-linear allocation policy trained end to end on net performance [12, 13, 14, 16, 17, 53]. Because an agent that maximises expected return has no explicit reason to avoid tail losses, a common remedy is to put risk terms, such as a rolling CVaR or a drawdown, into the reward [47, 49, 50]. Whether such terms make an allocator better or safer *out of sample, after realistic frictions, and beyond what simple portfolio rules achieve* is an empirical question, and the answer depends on evaluation details that are easy to get wrong: when orders are filled relative to the information used to make them, how training-seed variability enters the inference, and how many comparisons are made.

We study the most common member of this family, per-step penalties on the agent's own rolling CVaR and drawdown,

$$R_t = \log(1 + r_t) - \lambda_1 \widehat{\mathrm{CVaR}}_{\alpha,t} - \lambda_2 D_t^2,$$

in a simulator with commission, half-spread and square-root market impact [25, 26, 27], and ask:

> **Research question.** Under tradable execution, realistic frictions and strictly out-of-sample evaluation, do tail-risk reward penalties make PPO allocation agents better or safer, relative to an unpenalised agent, to classical allocation rules and to the network's own untrained starting point?

> **Main finding.** They do not make the agents better than 1/N, and they make them only as much safer as holding cash would. How a PPO allocator trades governs its net performance far more than how its reward is shaped: slowing execution is the only design change with a significant effect. And the untrained network's equal-weight-plus-cash prior is a benchmark no trained agent beats.

A correction. Versions 1–5 of this study (circulated as *Risk-Bounded Deep Reinforcement Learning for Dynamic Portfolio Allocation under Tail-Risk and Market Impact*) filled every order at the close of the same day whose close, range and volume the policy had just observed. That convention cannot be traded, and it inflated the fast-trading agents' results enough to create an apparent pre-cost timing edge over 1/N, which those versions reported as their most interesting positive finding. This version fixes the convention, reruns the decisive experiments with more seeds, and tests pre-specified hypotheses. The timing claim is withdrawn (§5, §6.4). We also drop "risk-bounded" from the title: fixed reward penalties carry no guarantee of bounded risk, and we found no evidence of one.

Our contributions are:

1. **An evaluation with tradable execution and pre-specified inference.** Decisions use data up to the close of day t; orders are share quantities fixed at that close and filled at the open of t+1. Five primary comparisons, the estimand (the expected out-of-sample Sharpe ratio of a newly trained agent), a two-level bootstrap over seeds and time, Holm correction, an equivalence margin and power calculations were committed before the rerun's results were inspected.
2. **A documented execution-timing correction**: what the same-close convention does, how much it inflates the same trained policies, and regression tests that make it detectable (§5).
3. **Controls that separate learning from portfolio construction**: an untrained-policy control, an exposure-matched (de-levered) 1/N, a decomposition of pre-cost performance into timing and static tilts, and regime-conditional de-risking diagnostics.

4. **Robustness checks**: costs from zero to three times the base case, no impact and tripled impact, an extra session of fill delay, an inherited starting book, warm-started versus from-scratch training, four independent simulated markets (one in which holding cash pays), and the frequency of reward clipping (§7).
5. **An open, tested framework** (Gymnasium environment, cost model, baselines, inference) with every configuration and output.

2 Related work

2.1 Classical portfolio construction

Markowitz [1] formalised the mean-variance trade-off; Black and Litterman [2] stabilised it with equilibrium returns; the Kelly criterion [3] maximises expected log growth, the return term of our reward. DeMiguel, Garlappi and Uppal [4] showed that the gains from optimisation are typically consumed by estimation error, making 1/N a demanding benchmark. Ledoit–Wolf shrinkage [5] regularises the covariance matrix; equal risk contribution [6, 7] and hierarchical risk parity [8] avoid expected returns. Coherent risk measures [11] and the linear-programming representation of CVaR [9, 10] made tail-risk optimisation tractable, and drawdown-constrained allocation has its own literature [36, 37]. Volatility-managed portfolios scale exposure inversely to recent variance [45].

2.2 Reinforcement learning for allocation

Moody et al. [12] and Moody and Saffell [13] trained trading systems by direct reinforcement on the differential Sharpe ratio; Almahdi and Yang [47] replaced it with a Calmar-ratio objective (return over expected maximum drawdown) for an ETF portfolio. Deep architectures followed [14, 15, 16], including state augmentation with return forecasts [48], market-condition embeddings that shift exposure using a drawdown-based risk measure [49], and rewards that explicitly charge transaction and risk costs [50]. Libraries such as FinRL [17] standardised PPO-based [18] environments, and several studies compare DRL with mean-variance optimisation directly [51, 52]. Hambly, Xu and Yang [53] survey the area. Table 1 places the closest studies against this one.

2.3 Risk-sensitive and constrained reinforcement learning

Constrained MDPs [24] add expected-cost constraints. Chow et al. [21] derived policy gradients for CVaR-constrained MDPs; Tamar et al. [20] estimated CVaR gradients by sampling; CPO [22] and RCPO [23] enforce constraints by trust regions and by adapting Lagrange multipliers on a slower timescale. Coache and Jaimungal [54] optimise time-consistent dynamic convex risk measures with an actor-critic method, and deep hedging [41] minimises convex risk measures of hedged P&L under frictions. Our penalty sits at the simple end of this spectrum: fixed multipliers on a rolling empirical CVaR of realised returns rather than a distributional quantity of the return-to-go. That simplicity is what most applied studies use, which is why we test it.

2.4 Transaction costs and market impact

Almgren and Chriss [25] separated temporary from permanent impact in optimal execution; empirical work supports a concave, roughly square-root dependence of impact on trade size relative to volume [26, 27]. Under quadratic costs, Gârleanu and Pedersen [46] show that the optimal policy

trades a constant fraction of the way towards an aim portfolio, the motivation for our smoothed-execution variants.

2.5 Evaluation methodology

Backtest overfitting inflates Sharpe ratios when many configurations are tried [33], and Arnott, Harvey and Markowitz [55] propose a research protocol, including prior hypotheses and full disclosure of trials, against it. Sharpe-ratio differences must account for serial dependence and cross-correlation [30]; we use circular block resampling [29]. In RL, results vary strongly across seeds, so interval estimates over seeds are needed [31, 32]. We combine both levels in one bootstrap, adjust the primary family with Holm's procedure [56], and test equivalence with two one-sided tests [57].

Table 1: The closest prior work. Entries summarise each paper's published abstract or record; they describe what each study sets out to do, not an assessment of its protocol.

Study	Setting	Risk in the objective	Costs	Relation to this study
Moody & Saffell 2001 [13]	Direct RL trading	Differential Sharpe ratio	Proportional	Origin of risk-adjusted RL objectives; we test additive tail penalties instead.
Almahdi & Yang 2017 [47]	Recurrent RL, ETF portfolio	Calmar ratio (expected maximum drawdown)	Several cost levels; stop-loss retraining	Closest drawdown-based design on ETFs; we add impact, next-open fills, seeds and controls.
Jiang, Xu & Liang 2017 [14]	Deep policy network, crypto	Log return	Commission in the reward	Return-only reference design (our variant A).
Ye et al. 2020 [48]	DRL with forecast-augmented state	Return	–	State design; we keep a fixed feature set and vary reward and execution.
Zhang, Zohren & Roberts 2020 [16]	DRL on futures	Volatility-scaled return	Proportional, varied	Shows costs drive net results; we add impact and fill timing.
Wang et al. 2021 [49]	DRL with market-condition unit	Drawdown-based risk; long/short split	–	Aims at state-dependent de-risking, which our agents do not learn.
Zhang et al. 2022 [50]	Two-stream policy network	Reward constrains risk cost	Transaction cost in the reward	Closest cost- and risk-aware reward; we test fixed penalties against 1/N with seeds.
Benhamou et al. 2020 [51]; Sood et al. 2023 [52]	DRL vs mean-variance	Various	–	Report DRL competitive with or better than MVO; we also include 1/N, risk parity and an untrained-policy control.
Tessler et al. 2019 [28]; Chow et al. 2018 [21]	Constrained / CVaR RL	Lagrangian constraints, CVaR	n/a	Methods behind our adaptive-multiplier variants (legacy runs, §8).
Coache & Jaimungal 2024 [54]	Risk-sensitive actor-critic	Dynamic convex risk measures	Problem-specific	The principled alternative to rolling penalties, not tested here (§10).

3 Problem formulation

3.1 Markov decision process

Time is indexed by trading days t. The investor holds N risky assets and cash, with net asset value NAV_t marked at the close, share holdings $h_t \in \mathbb{R}^N$, close prices P_t and open prices O_t. Decisions are taken after the close of t using information available at that close.

State. The observation is

$$s_t = \big(X_{t-L+1:t},\ w_t^-,\ w_t^{\mathrm{cash}},\ D_t,\ \hat{\sigma}_t^p \big), \qquad X \in \mathbb{R}^{L\times N\times F},$$

where X holds $F = 20$ causally normalised features per asset over $L = 20$ days (§4.3), w_t^- are the current (drifted) weights, $D_t = 1 - \mathrm{NAV}_t / \max_{\tau\le t} \mathrm{NAV}_\tau$ is the running drawdown, and $\hat{\sigma}_t^p$ is the 21-day realised portfolio volatility. Entries are clipped to $[-10, 10]$; the observation has $LNF + N + 3$ dimensions (4,013 for ten assets).

Action. The policy outputs $a_t \in [-1, 1]^{N+1}$, mapped to target weights by a temperature-scaled softmax with an explicit cash logit,

$$(w_t^\star, w_t^{\mathrm{cash}\star}) = \mathrm{softmax}(\kappa\, a_t), \qquad \kappa = 3,$$

so the agent is long-only and fully funded and can de-risk continuously into cash. At initialisation the actor's mean action is close to zero, so the untrained policy holds about $1/(N+1)$ in every asset and in cash.

3.2 Execution timing and friction-adjusted returns

Timeline. At the close of t the agent observes s_t and chooses target weights. The order is a vector of *share quantities* fixed at that moment from information up to the close of t. It is filled at the open of $t+1$; the existing book is carried overnight, so it earns (or loses) the overnight gap O_{t+1}/P_t.

Costs are charged on the executed notional at the open, using volatility and average daily volume (ADV) known at t.

Order sizing. With decision-time NAV and weights $w_t^- = h_t P_t/\mathrm{NAV}_t$, the order is

$$n_t = \frac{\mathrm{clip}(\tilde{w}_t\,\widetilde{\mathrm{NAV}}_t - h_t P_t,\ -\rho\,\mathrm{ADV}_t,\ \rho\,\mathrm{ADV}_t)}{P_t},$$

where $\widetilde{\mathrm{NAV}}_t$ is the NAV net of the cost of the trade itself, found by a short fixed-point iteration at decision-time prices, $\rho = 10\%$ caps participation, and $\tilde{w}_t = w_t^\star$ except under smoothed execution (below). The executed notional is $q_t = n_t \odot O_{t+1}$. Because the order is fixed in shares, an overnight gap moves the executed notional slightly away from the target; for 1/N this leaves at most 0.5% of NAV as negative cash on typical days (2.5% in the worst case), financed at the zero cash rate.

Costs. With $\sigma_{i,t}$ the 20-day daily volatility and $\mathrm{ADV}_{i,t}$ the 20-day mean dollar volume (both up to t), the per-unit square-root impact is $I_{i,t} = \sigma_{i,t}(|q_{i,t}|/\mathrm{ADV}_{i,t})^{1/2}$, and the dollar cost is

$$C_t = \sum_{i=1}^{N}\Big[\ \underbrace{(c+s)\,|q_{i,t}|}_{\text{commission + half-spread}} + \underbrace{\eta\, I_{i,t}\,|q_{i,t}|}_{\text{temporary}} + \underbrace{\tfrac{1}{2}\gamma\, I_{i,t}\,|q_{i,t}|}_{\text{permanent, own trade}} + \underbrace{\psi_{i,t}\,q_{i,t}}_{\text{residual displacement}}\ \Big],$$

where permanent impact is a per-asset displacement state that each trade pushes in its own direction and that decays with half-life h: $\psi_{i,t+1} = \delta(\psi_{i,t} + \gamma I_{i,t}\,\mathrm{sign}(q_{i,t}))$, $\delta = 2^{-1/h}$. Positions are marked at historical, un-impacted prices, so impact never creates paper gains; a round trip on a flat price path always loses money (tested).

Accounting.

$$\mathrm{cash}_t^+ = \mathrm{cash}_t - n_t^\top O_{t+1} - C_t, \qquad h_t^+ = h_t + n_t, \qquad \mathrm{NAV}_{t+1} = \mathrm{cash}_t^+ + h_t^{+\top} P_{t+1}, \qquad r_t = \frac{\mathrm{NAV}_{t+1}}{\mathrm{NAV}_t} - 1.$$

Gross-of-cost returns add C_t/NAV_t back to r_t. Parameters: $c = 1$ bp, $s = 2$ bp, $\eta = 0.15$, $\gamma = 0.05$, $h = 5$ days, $\rho = 10\%$, initial NAV \$10m per fold. On days when a rule does not rebalance it sends no order at all, so its book simply drifts at zero cost.

Smoothed execution (variants S and SE). The weights actually targeted are a partial adjustment from the current weights towards the policy's target, with a no-trade band b:

$$\tilde{w}_t = w_t^- + \kappa_{\mathrm{tr}}\,(w_t^\star - w_t^-), \qquad \tilde{w}_t \leftarrow w_t^- \ \text{ if } \ \|\tilde{w}_t - w_t^-\|_1 < b,$$

with $\kappa_{\mathrm{tr}} = 0.1$ (a half-life of about a week) and $b = 1\%$. The policy supplies the aim portfolio and the environment applies a fixed trading speed [46].

3.3 Tail-risk measures

For losses $\ell = -r$, CVaR at level α is the expected loss in the worst $1-\alpha$ tail [43]. We use the Rockafellar–Uryasev representation [9, 10], exact for empirical distributions:

$$\widehat{\mathrm{CVaR}}_{\alpha,t} = \widehat{\mathrm{VaR}}_{\alpha,t} + \frac{1}{(1-\alpha)W}\sum_{k=0}^{W-1}\big(\ell_{t-k} - \widehat{\mathrm{VaR}}_{\alpha,t}\big)^+,$$

with $\alpha = 0.95$, a rolling window of $W = 63$ realised returns and at least 20 observations before the penalty switches on.

3.4 Risk-penalised reward

$$R_t = k \cdot \mathrm{clip}_{[-10/k,\,10/k]}\Big(\log(1+r_t) \;-\; \lambda_1\big[\widehat{\mathrm{CVaR}}_{\alpha,t}\big]^{+} \;-\; \lambda_2\, D_{t+1}^2\Big), \qquad k = 100,$$

and PPO maximises $J(\pi) = \mathbb{E}_\pi\big[\sum_t \gamma^t R_t\big]$. J is the Lagrangian of a constrained MDP evaluated at fixed multipliers: nothing sets a target risk level, so the penalties move the optimum along the return/risk frontier by an amount that depends on their scale, and no bound on risk follows. The CVaR term depends on 63 days of the agent's own returns, which the state summarises only through D_t and $\hat{\sigma}_t^p$, so the reward is not Markov in s_t. In daily data the return term is small (about 3×10^{-4} for 20% volatility, against a daily CVaR of about 2.5×10^{-2}), so we use multipliers scaled to daily magnitudes, $(\lambda_1, \lambda_2) = (0.05, 0.5)$ (variant E); the legacy runs also used $\lambda \in \{0, 1\}$ (§8). The clip rarely binds (§7.4).

3.5 Policy architecture and optimisation

Each asset's $L \times F$ history passes through a shared causal temporal convolutional encoder [38] (two dilated layers, kernel 3, width 32, receptive field 7 days); a learned asset embedding is added, one pre-norm transformer block [39] (4 heads) mixes information across assets, and the result is fused with an embedding of the portfolio state into a 128-dimensional vector shared by a Gaussian actor (state-independent log-std, initialised at -0.5) and a critic, each with hidden layers [128, 64] and tanh activations. Agents are trained with PPO [18] and generalised advantage estimation [19] in Stable-Baselines3 [44] with the hyper-parameters in Table 4. Rewards are normalised by a running estimate of the return scale; observations are not, because the features are already z-scores. Training episodes are 252 days long, start at random dates inside the training window, begin in cash, and use the same next-open execution as evaluation.

4 Empirical setup

4.1 Market data

The study uses ten liquid U.S.-listed ETFs covering the major asset classes: U.S., developed ex-U.S. and emerging equities (SPY, EFA, EEM); intermediate and long Treasuries (IEF, TLT); investment-grade and high-yield credit (LQD, HYG); gold (GLD); broad commodities (DBC); and U.S. real estate (VNQ). Daily open, high, low, close and volume come from Yahoo Finance, adjusted for splits and distributions. The sample runs from 11 April 2007 (HYG's first trading day) to 14 October 2026: 4,910 trading days with no gaps or zero-volume days. The raw file is not redistributed; `scripts/download_data.py` rebuilds it and records its provenance, and the release manifest records its SHA-256 hash (§12).

Table 2: The ETF universe, April 2007 – October 2026 (4,910 days). Daily log returns, annualised. "Stress" days fall in the named episodes of Table 3. *Note:* Average pairwise correlation 0.27 on calm days and 0.21 on stress days; SPY–TLT correlation -0.24 calm and -0.42 stress (flight to quality).

ETF	Asset class	Ann. mean (%)	Ann. vol (%)	Vol, calm (%)	Vol, stress (%)	Skew	Excess kurt.	Worst day (%)	Mean ADV ($bn)
SPY	U.S. equity	10.43	19.63	14.75	33.94	−0.29	13.95	−11.59	24.27
EFA	Developed ex-U.S. equity	4.41	21.65	16.69	36.69	−0.34	13.24	−11.84	1.03
EEM	Emerging equity	4.49	27.91	20.76	49.14	0.05	16.27	−17.63	1.80
IEF	7–10y Treasuries	2.98	6.94	6.25	9.56	0.11	2.57	−2.54	0.34
TLT	20y+ Treasuries	2.60	15.08	13.67	20.52	0.01	3.17	−6.90	1.25
LQD	IG corporate credit	3.77	8.81	6.48	15.73	−0.44	56.47	−9.55	0.94
HYG	High-yield credit	4.64	10.79	7.80	19.38	0.32	40.21	−8.44	1.06
GLD	Gold	8.85	18.22	16.64	24.42	−0.42	7.00	−10.84	1.64
DBC	Commodities	2.46	19.33	16.57	29.02	−0.50	3.39	−8.28	0.04
VNQ	U.S. real estate	4.84	29.56	21.82	52.30	−0.52	18.00	−21.71	0.23

Stress episodes. Real data has no ground-truth regimes. For regime-conditional analysis only, we label days in ten widely recognised stress episodes (Table 3); they cover 17.3% of the sample and 11.1% (391) of the test days. No strategy observes these labels. The episodes are dated with hindsight.

Table 3: Named stress episodes used to label regimes (inclusive dates; S&P 500 peak to trough unless noted).

Episode	From	To	In test period?
Global financial crisis	2007-10-09	2009-03-09	no (training only)
Flash crash / euro sovereign stress	2010-04-23	2010-07-02	no
U.S. downgrade / euro crisis	2011-07-22	2011-10-03	no
China devaluation sell-off	2015-08-18	2015-09-29	yes
Oil / China sell-off	2015-12-29	2016-02-11	yes
Volatility shock	2018-01-26	2018-02-08	yes
Q4 2018 sell-off	2018-09-21	2018-12-24	yes
COVID-19 crash	2020-02-19	2020-03-23	yes
2022 inflation / rates bear market (stocks and bonds)	2022-01-03	2022-10-12	yes
2025 tariff shock	2025-02-19	2025-04-08	yes

4.2 Simulated markets

Simulated markets provide what real data cannot: ground-truth regimes and independent replications. A two-state Markov regime in the spirit of Hamilton [40] drives five assets: calm turns to stress with probability 0.008 per day and stress persists with probability 0.97. Assets follow a one-factor model with Student-t innovations ($\nu = 4$); in stress, volatility rises 2.5-fold, factor loadings rise from 0.55 to 0.85 and drift falls to -35% a year. Calm-regime drifts are drawn between 2% and 10% a year, and each asset carries a weak, persistent latent alpha (AR(1) coefficient 0.97, scale 0.06 daily standard deviations). Opens are the previous close times a random overnight gap. Each market has 3,000 days. For the robustness study (§7.3) we draw three new standard markets and one **bear market** with calm-regime drifts between -8% and 0% a year and stress twice as frequent, a setting in which holding cash pays and a tail-risk-aware agent should beat 1/N. The legacy controlled study (§8) used a single market (appendix Table 27).

4.3 Features

For each asset and window $w \in \{5, 21, 63\}$ we compute log momentum, realised volatility, distance from the moving average, log relative volume, and two order-flow-imbalance proxies: bulk-volume classification [28] and close-location-value-weighted volume. The one-day return and the log high/low range complete the set ($F = 20$). Every feature is z-scored with a rolling 252-day window using data up to and including t and clipped to ± 5. Features use the close, range and volume of day t, all known at the decision time. Unit tests confirm that perturbing any data after t leaves every feature, the observation, the target weights and the order at t unchanged, and that features are invariant to rescaling the price history (so later dividend or split adjustments cannot leak in).

4.4 Walk-forward protocol

We use 14 rolling, end-anchored folds, each with a 1,008-day (four-year) training window, a 5-day embargo and a 252-day test window. The test windows form one contiguous track of 3,527 daily returns from 2 October 2012 to 14 October 2026 (fold calendar in Table 20). Each (variant, seed)

is trained as a chain: fold 0 trains from scratch for 102,400 steps, and each later fold fine-tunes the previous fold's model on its own training window for 28,672 steps. Only data before each test window is used for training. Each fold is evaluated with its final model, with no checkpoint selection. §7.2 compares warm-started with from-scratch training. The simulated markets use the same chain protocol with four folds.

4.5 Strategies

PPO variants (5 seeds each) share every hyper-parameter (Table 4):

- **A**: return only ($\lambda_1 = \lambda_2 = 0$);
- **E**: calibrated CVaR and drawdown penalties ($\lambda_1 = 0.05$, $\lambda_2 = 0.5$);
- **S**: return only, with smoothed execution ($\kappa_{\text{tr}} = 0.1$, $b = 1\%$);
- **SE**: smoothed execution with E's penalties.

These four were chosen, before the rerun, as the variants that decide the paper's claims. Variants that appear only in the legacy runs (CVaR-only and drawdown-only penalties at $\lambda = 1$, a turnover penalty, RCPO and a volatility overlay) are summarised in §8.

Classical baselines, long-only and fully invested, on trailing 252-day estimates and executed through the same simulator: 1/N rebalanced monthly (every 21 days) and daily, and 1/N buy-and-hold [4]; mean-variance [1] with Ledoit–Wolf covariance [5], risk aversion 5 and sample means shrunk halfway to their cross-sectional average; equal risk contribution [6, 7], monthly and daily; and hierarchical risk parity [8].

Controls. The *untrained policy* holds $1/(N+1)$ in every asset and in cash, rebalanced daily: PPO's starting point, which shows what training adds. The *exposure-matched control* scales 1/N by each agent's average invested fraction, holding the rest in cash, which shows what an agent's cash buffer alone is worth.

4.6 Metrics and pre-specified inference

We report cumulative and annualised return, volatility, the Sharpe ratio (zero risk-free rate), the Sortino ratio [34], maximum drawdown, the Calmar ratio [35], daily CVaR_{95}, turnover and costs, computed on each seed's stitched test track. The analysis plan (`ANALYSIS_PLAN.md`) was committed before any rerun result was inspected and fixes:

- **Estimand.** For a PPO variant, the expected out-of-sample metric of a newly trained agent, estimated as the mean of per-seed metrics. Baselines are deterministic.
- **Two-level bootstrap.** Each replication resamples each variant's seeds with replacement, then resamples days in circular 21-day blocks with the same indices for every series, and recomputes the difference (2,000 replications; percentile intervals; two-sided p-values from the bootstrap distribution centred under the null).
- **Primary family** (net Sharpe unless stated; Holm-adjusted at $\alpha = 0.05$): P1 E − A (do penalties change performance?); P2 S − A (does smoothing?); P3 SE − 1/N (does the best-designed agent differ from 1/N?); P4 A − 1/N; P5 gross A − gross 1/N (is there a pre-cost timing edge?).
- **Equivalence.** Two strategies are equivalent only if the 90% interval lies inside ±0.15 Sharpe (TOST at $\alpha = 0.05$); a non-significant difference that fails this test is reported as inconclusive, never as parity.
- **Power.** The minimum detectable effect at 80% power, MDE = 2.80 × bootstrap SE.

- **Decision rule.** If P5 is not significant after correction, the timing claim of versions 1–5 is withdrawn.

All other comparisons are secondary and uncorrected.

Table 4: PPO hyper-parameters, shared by every variant.

Parameter	Value	Parameter	Value
Parallel envs × rollout length	8 × 512 (4,096 per update)	Minibatch / epochs	256 / 10
Learning rate	3×10^{-4}, linear decay per training call	Discount γ / GAE λ	0.99 / 0.95
Clip range / target KL	0.2 / 0.03	Value / entropy coefficient	0.5 / 0
Max grad-norm	0.5	Reward normalisation	running return std
Episode length	252 days, random start, from cash	Seeds	5 (ETF); 2 (simulated markets)
Budget	102,400 steps (fold 0) + 28,672 per later fold	Execution	decide at close t, fill at open t+1

5 The execution-timing correction

5.1 What was wrong

Versions 1–5 of this study computed features from the close, range and volume of day t and then filled the resulting order at the close of the same day t. A real investor cannot see the closing price and also trade at it: the decision would have to be made before the information it uses exists. The error matters most for agents that trade a lot on short-horizon features. A policy that reacts to the day's last move at the price where that move ended can capture short-term reversals from the close onwards, part of which, in practice, happens overnight before any order can be filled. Slow strategies are hardly affected: the monthly baselines changed by at most 0.012 Sharpe between conventions.

5.2 How it was fixed and verified

The simulator now separates the decision from the fill (§3.2): orders are share quantities fixed at the decision close and filled at the next open, the old book is carried overnight, and cost inputs are known at the decision. A first rerun was stopped after about 30 minutes, before any result was inspected, when a further audit found that the order had been converted from weights into shares at the open of $t + 1$; the analysis plan was amended accordingly before restarting (Amendment 1). The test suite now checks, among other things:

- that the observation, the target weights *and the order* at t are unchanged when every price, volume and open from $t + 1$ onwards is perturbed;
- a two-asset, three-day example whose NAV path, cash, holdings and costs are computed by hand;
- that existing positions earn the overnight gap and costs are charged on executed quantities at the fill price;
- that cash and weights stay consistent (weights plus cash sum to one; long-only books never go short) under large random gaps;
- that gross return equals net return plus cost, and that a buy-and-hold book never re-trades.

The look-ahead checklist in the analysis plan lists every information pathway (features, order generation, fill, cost inputs, fold boundaries, reward normalisation, baselines, price adjustments) and how each is controlled.

5.3 How much the convention mattered

Table 5: The execution-timing correction. Same data, folds, test dates, features, frictions and PPO settings; only the fill convention differs. Same-close (v1–v5, not tradable): decide and fill at the close of t. Next-open (this version): decide after the close of t, fill at the open of t+1. Same-close PPO rows are from 3 seeds, next-open rows from 5 seeds (new training runs).

Strategy	Net Sharpe, same-close	Net Sharpe, next-open	Change	Gross Sharpe, same-close	Gross Sharpe, next-open	Turnover %/day, same-close	Turnover %/day, next-open
PPO-A · return only (λ_1=0, λ_2=0)	0.38	0.11	−0.27	0.81	0.56	49.2	48.5
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.37	0.17	−0.19	0.78	0.60	40.6	40.4
PPO-S · smoothed execution (λ_1=0, λ_2=0)	0.59	0.60	+0.00	0.64	0.64	4.7	4.5
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.60	0.56	−0.04	0.64	0.60	3.9	3.9
Equal Weight (1/N)	0.65	0.64	−0.00	0.65	0.65	0.5	0.5
Buy & Hold (1/N)	0.66	0.65	−0.01	0.66	0.66	0.4	0.4
MVO (Ledoit-Wolf)	0.75	0.74	−0.01	0.77	0.76	2.8	2.8
Risk Parity (ERC)	0.60	0.60	−0.00	0.61	0.61	0.6	0.6
HRP	0.46	0.47	+0.00	0.48	0.48	0.9	0.9
Untrained policy (1/(N+1) incl. cash)	0.66	0.64	−0.01	0.66	0.65	0.8	0.8

Table 5 compares the two conventions on the same data, folds and test dates. The fast-trading agents lose most of their performance: vanilla A falls from a net Sharpe of 0.38 to 0.11 and E from 0.37 to 0.17, and their gross Sharpe falls from above 1/N's to below it. The smoothed agents, which trade about 4% of the book a day, barely change (S 0.59 → 0.60, SE 0.60 → 0.56), and neither do the baselines. The size of the artifact can also be measured without retraining: evaluating the *next-open-trained* policies under the same-close fill raises vanilla A's Sharpe ratio from 0.11 to 0.27 and S's from 0.60 to 0.61 (Table 15). The convention flatters fast traders, whatever they were trained on.

6 Results under tradable execution

6.1 Overview

Table 6: Out-of-sample performance on the stitched walk-forward test period (14 folds, 3,527 trading days, net of all frictions). PPO rows: mean ± standard deviation across 5 training seeds. Bold marks the best value in each column. *Note:* CVaR_{95} is the daily historical expected shortfall of net returns. Annualised return is geometric; Sharpe and Sortino use a zero risk-free rate.

Strategy	Cum. return (%)	Ann. return (%)	Ann. vol (%)	Sharpe	Sortino	MaxDD (%)	Calmar	CVaR_{95} (%)
PPO-A · return only (λ_1=0, λ_2=0)	8.5 ± 5.7	0.6 ± 0.4	9.8 ± 0.7	0.11 ± 0.04	0.15 ± 0.05	28.2 ± 1.4	0.02 ± 0.01	1.46 ± 0.11
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	16.3 ± 8.0	1.1 ± 0.5	8.3 ± 0.8	0.17 ± 0.06	0.24 ± 0.09	22.1 ± 1.3	0.05 ± 0.02	1.26 ± 0.11
PPO-S · smoothed execution (λ_1=0, λ_2=0)	95.3 ± 22.4	4.9 ± 0.9	8.5 ± 0.4	0.60 ± 0.10	0.83 ± 0.14	19.5 ± 1.1	0.25 ± 0.05	1.27 ± 0.05
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	75.8 ± 16.0	4.1 ± 0.7	7.6 ± 0.2	0.56 ± 0.09	0.78 ± 0.13	19.1 ± 1.7	0.22 ± 0.05	1.14 ± 0.03
Equal Weight (1/N)	108.3	5.4	8.8	0.64	0.89	19.7	0.27	1.29
Buy & Hold (1/N)	111.0	5.5	8.8	0.65	0.90	19.9	0.28	1.30
MVO (Ledoit-Wolf)	**260.7**	**9.6**	13.6	**0.74**	**1.01**	24.5	**0.39**	2.15
Risk Parity (ERC)	74.5	4.1	7.0	0.60	0.83	19.5	0.21	1.06
HRP	49.1	2.9	**6.6**	0.47	0.65	20.5	0.14	**0.98**
Equal Weight (1/N, daily)	110.1	5.4	8.9	0.64	0.89	20.1	0.27	1.30
Risk Parity (ERC, daily)	74.5	4.1	7.0	0.60	0.84	19.8	0.21	1.05
Untrained policy (1/(N+1) incl. cash)	97.2	5.0	8.1	0.64	0.89	**18.4**	0.27	1.19

Over 14 years, 1/N earns a Sharpe ratio of 0.64 (monthly and daily rebalancing alike), buy-and-hold 0.65 and the untrained policy 0.64 with the smallest maximum drawdown of any strategy (18.4%). Ledoit–Wolf mean-variance has the highest Sharpe ratio (0.74), with a 24.5% drawdown, and is not distinguishable from 1/N. Among the agents, the unsmoothed ones are far behind (A 0.11, E 0.17) and the smoothed ones close to the classical rules (S 0.60, SE 0.56). Individual seeds are in appendix Table 22: the Sharpe ratio of S ranges from 0.43 to 0.67 across seeds, and no seed of any variant exceeds 1/N by more than 0.03.

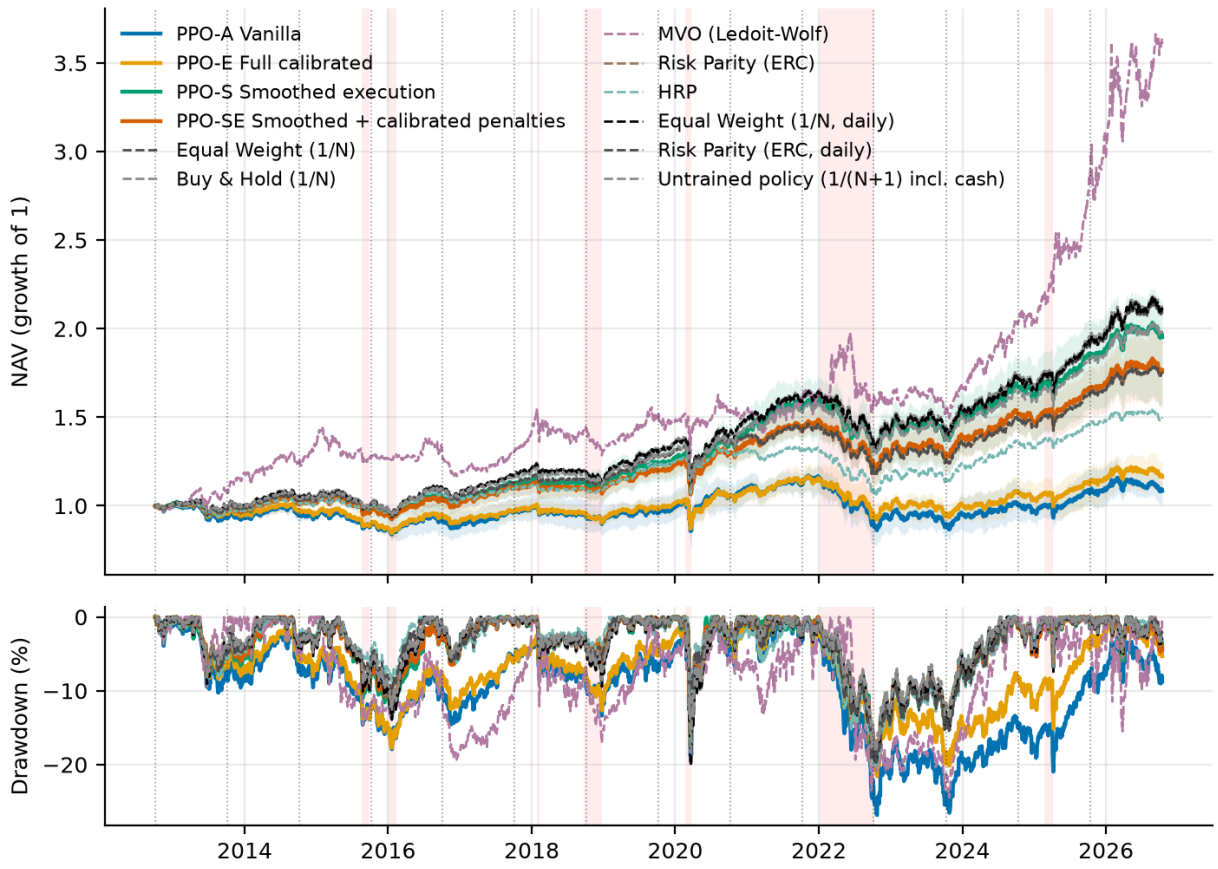


Figure 1: Out-of-sample NAV (top) and drawdown (bottom) under next-open execution, net of all frictions. PPO lines are seed means with min–max bands over 5 seeds; red shading marks the named stress episodes; dotted lines mark the 14 fold boundaries.

Table 7: Pre-specified primary comparisons (ANALYSIS_PLAN.md, committed before the rerun's results were seen). Two-level bootstrap: training seeds resampled within each strategy, then days resampled in circular 21-day blocks shared by all series (2,000 replications). The estimand is the expected Sharpe ratio of a newly trained agent (mean of per-seed Sharpe ratios). *Note:* Holm adjustment over the five primary tests (family-wise $\alpha = 0.05$). Equivalence: TOST with margin ±0.15 Sharpe, i.e. the 90% CI inside (−0.15, 0.15). "Inconclusive" means neither a significant difference nor equivalence. MDE_{80}: smallest true difference detectable with 80% power, 2.80 × bootstrap SE.

ID	Comparison (Sharpe)	Estimate [95% CI]	90% CI	MDE_{80}	p	p (Holm)	Verdict
P1	PPO-E − PPO-A	+0.064 [−0.119, 0.260]	[−0.078, 0.221]	0.26	0.482	0.880	not significant; inconclusive
P2	PPO-S − PPO-A	+0.490 [0.271, 0.735]	[0.304, 0.698]	0.34	<0.001	<0.001	significant; different
P3	PPO-SE − 1/N	−0.079 [−0.234, 0.062]	[−0.207, 0.040]	0.21	0.293	0.880	not significant; inconclusive
P4	PPO-A − 1/N	−0.533 [−0.788, −0.316]	[−0.742, −0.345]	0.33	<0.001	<0.001	significant; different
P5	PPO-A − 1/N (gross of costs)	−0.088 [−0.304, 0.116]	[−0.268, 0.086]	0.30	0.396	0.880	not significant; inconclusive

6.2 Pre-specified tests

P1. Penalties: no detectable effect on Sharpe. E − A = +0.06 (95% CI [−0.12, 0.26], Holm p = 0.88). The interval also exceeds the equivalence margin, so the result is inconclusive; effects smaller than about 0.26 Sharpe could not be detected.

P2. Smoothed execution: a large, significant improvement. S − A = +0.49 ([0.27, 0.74], Holm p < 0.001). This is the only design choice with a significant effect on risk-adjusted return.

P3. The best-designed agent versus 1/N: inconclusive. SE − 1/N = −0.08 ([−0.23, 0.06], Holm p = 0.88). SE is not significantly worse than 1/N, but its 90% interval [−0.21, 0.04] is not inside ±0.15, so it is not shown to be equivalent either. Versions 1–5 described the smoothed agents as "at parity" with 1/N; that wording was not justified.

P4. Vanilla PPO is significantly worse than 1/N. A − 1/N = −0.53 ([−0.79, −0.32], Holm p < 0.001).

P5. No pre-cost timing edge. Gross A − gross 1/N = −0.09 ([−0.30, 0.12], Holm p = 0.88). Under the decision rule, the timing claim of versions 1–5 is withdrawn.

Two primary tests are significant, and both concern execution: smoothing helps, and unsmoothed PPO loses to 1/N. Neither involves the tail-risk penalties.

6.3 What the penalties do change

Table 8: Secondary comparisons (not used for claims; uncorrected): differences in other metrics with 95% two-level bootstrap intervals. *Note:* * p < 0.05, ** p < 0.01, uncorrected. Negative MaxDD / CVaR differences mean the first strategy has less tail risk. The bootstrap distribution of a maximum-drawdown difference is strongly skewed, so its percentile interval and its centred p-value can disagree; both are reported as computed.

Comparison	Ann. return (pp)	MaxDD (pp)	$CVaR_{95}$ (pp)	Turnover (pp/day)
PPO-E − PPO-A	+0.50 [−1.21, 2.21]	−6.1 [−13.4, 3.1]	−0.20 [−0.36, −0.07]**	−8.1 [−10.1, −6.2]**
PPO-S − PPO-A	+4.28 [2.40, 6.27]**	−8.7 [−25.4, −1.9]	−0.20 [−0.33, −0.10]**	−44.0 [−45.4, −42.7]**
PPO-SE − PPO-E	+3.01 [1.63, 4.49]**	−3.1 [−19.3, −0.8]	−0.12 [−0.23, −0.01]*	−36.6 [−38.3, −34.7]**
PPO-S − 1/N	−0.53 [−1.92, 0.73]	−0.3 [−4.9, 3.6]	−0.03 [−0.09, 0.03]	+4.0 [3.8, 4.2]**
PPO-SE − 1/N	−1.29 [−2.67, 0.05]	−0.7 [−6.5, 2.7]	−0.15 [−0.22, −0.09]**	+3.4 [3.2, 3.5]**
PPO-SE − ERC	+0.03 [−1.17, 1.21]	−0.5 [−3.0, 5.3]	+0.09 [0.04, 0.14]**	+3.3 [3.1, 3.5]**
PPO-SE − Untrained policy	−0.88 [−2.07, 0.30]	+0.7 [−4.5, 4.4]	−0.04 [−0.11, 0.01]	+3.0 [2.8, 3.2]**

The penalties do change the agents' risk profile (Table 8, secondary and uncorrected). Relative to A, E lowers daily $CVaR_{95}$ by 0.20 percentage points ([0.07, 0.36]) and turnover by 8 points a day, and its maximum drawdown is 6 points lower (22.1% against 28.2%; interval [−13.4, +3.1]). Relative to 1/N, SE has a lower daily CVaR (by 0.15 points, [0.09, 0.22]) and a slightly lower annual return (−1.3 points, [−2.7, +0.1]). These reductions are what holding more cash produces (§6.5): SE holds 15% cash on average and E 10%, against 7% for A and none for 1/N.

Costs explain most of the gap between unsmoothed and smoothed agents. A and E trade 40–49% of the book every day and pay 50–62% of NAV in costs over 14 years, a drag of 3.6–4.4% a year. The smoothed agents trade 3.9–4.5% a day and pay 4.4–5.2% of NAV, still seven to eight times the cost of monthly 1/N.

6.4 No timing edge before costs

Before costs no agent beats 1/N (gross Sharpe 0.56–0.64 against 0.65; all intervals include zero). Holding each agent's average weights statically would have earned 0.65–0.67 gross, more than the

Table 9: Trading activity, transaction costs and allocation behaviour over the 3,527-day out-of-sample period. Each fold starts from cash with a $10m NAV. *Note:* Costs sum commission, half-spread, temporary and permanent impact, each period expressed as a fraction of pre-trade NAV. Effective N = $(\Sigma|w|)^2$ / Σw^2.

Strategy	Turnover (%/day)	Turnover (×/yr)	Total cost (% NAV)	Cost drag (%/yr)	Avg gross exposure (%)	Effective N	Max weight (%)
PPO-A · return only (λ_1=0, λ_2=0)	48.5 ± 1.3	122 ± 3	61.6 ± 1.6	4.40 ± 0.11	92.6	5.75	74.4
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	40.4 ± 2.0	102 ± 5	50.0 ± 2.7	3.58 ± 0.19	89.7	6.48	68.7
PPO-S · smoothed execution (λ_1=0, λ_2=0)	4.5 ± 0.1	11 ± 0	5.2 ± 0.2	0.37 ± 0.01	89.4	7.71	44.7
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	3.9 ± 0.1	10 ± 0	4.4 ± 0.1	0.31 ± 0.01	85.1	7.88	40.6
Equal Weight (1/N)	0.5	1	0.6	0.05	100.0	9.99	12.5
Buy & Hold (1/N)	0.4	1	0.5	0.04	99.9	9.93	15.6
MVO (Ledoit-Wolf)	2.8	7	3.9	0.28	99.2	1.94	101.4
Risk Parity (ERC)	0.6	1	0.7	0.05	100.0	8.13	31.1
HRP	0.9	2	1.1	0.08	100.0	5.38	54.2
Equal Weight (1/N, daily)	0.9	2	1.1	0.08	100.0	10.00	11.3
Risk Parity (ERC, daily)	1.0	3	1.2	0.08	100.0	8.13	31.3
Untrained policy (1/(N+1) incl. cash)	0.8	2	1.0	0.07	90.9	10.00	10.2

Table 10: Performance before transaction costs, and co-movement with the 1/N portfolio. Gross returns add each period's cost back to the net return. Tests are paired block bootstraps on seed-averaged gross returns. *Note:* * $p < 0.05$, ** $p < 0.01$, uncorrected.

Strategy	Net Sharpe	Gross Sharpe	Gross ΔSharpe vs 1/N [95% CI]	Corr. with 1/N	Beta to 1/N	Tracking error vs 1/N (%)
PPO-A · return only (λ_1=0, λ_2=0)	0.11	0.56 ± 0.06	−0.033 [−0.216, 0.135] p=0.707	0.84	0.94	5.3
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.17	0.60 ± 0.07	−0.006 [−0.130, 0.113] p=0.923	0.91	0.86	3.8
PPO-S · smoothed execution (λ_1=0, λ_2=0)	0.60	0.64 ± 0.10	+0.010 [−0.102, 0.122] p=0.862	0.95	0.92	2.7
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.56	0.60 ± 0.09	−0.027 [−0.144, 0.084] p=0.651	0.95	0.82	2.9
Equal Weight (1/N)	0.64	0.65	–	1.00	1.00	0.0
Buy & Hold (1/N)	0.65	0.66	+0.010 [−0.023, 0.038] p=0.505	1.00	1.00	0.5
MVO (Ledoit-Wolf)	0.74	0.76	+0.115 [−0.371, 0.578] p=0.649	0.56	0.86	11.4
Risk Parity (ERC)	0.60	0.61	−0.040 [−0.171, 0.090] p=0.551	0.96	0.77	2.9
HRP	0.47	0.48	−0.168 [−0.383, 0.044] p=0.136	0.89	0.67	4.2
Equal Weight (1/N, daily)	0.64	0.65	+0.004 [−0.006, 0.016] p=0.543	1.00	1.01	0.3
Risk Parity (ERC, daily)	0.60	0.61	−0.033 [−0.166, 0.099] p=0.645	0.95	0.76	3.0
Untrained policy (1/(N+1) incl. cash)	0.64	0.65	+0.003 [−0.007, 0.016] p=0.591	1.00	0.92	0.8

Table 11: Timing versus static tilts, before costs. The static-tilt portfolio holds each variant's average out-of-sample weights constantly (rebalanced daily, no costs); the difference to the actual policy's gross Sharpe is the contribution of day-to-day re-allocation.

Strategy	Static tilt (gross Sharpe)	Actual policy (gross Sharpe)	Timing contribution	Avg cash (%)
PPO-A · return only (λ_1=0, λ_2=0)	0.65	0.56	−0.09	7.4
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.65	0.60	−0.05	10.3
PPO-S · smoothed execution (λ_1=0, λ_2=0)	0.66	0.64	−0.02	10.6
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.67	0.60	−0.06	14.9
1/N (static, daily rebalanced)	0.66	–	–	–

agents' actual gross Sharpe ratios, so day-to-day re-allocation *subtracted* 0.02–0.09 Sharpe before any cost was paid (Table 11). Under the same-close convention, the same decomposition had credited +0.04 to +0.17 to timing. The signal the earlier versions attributed to the agents came from the fill convention.

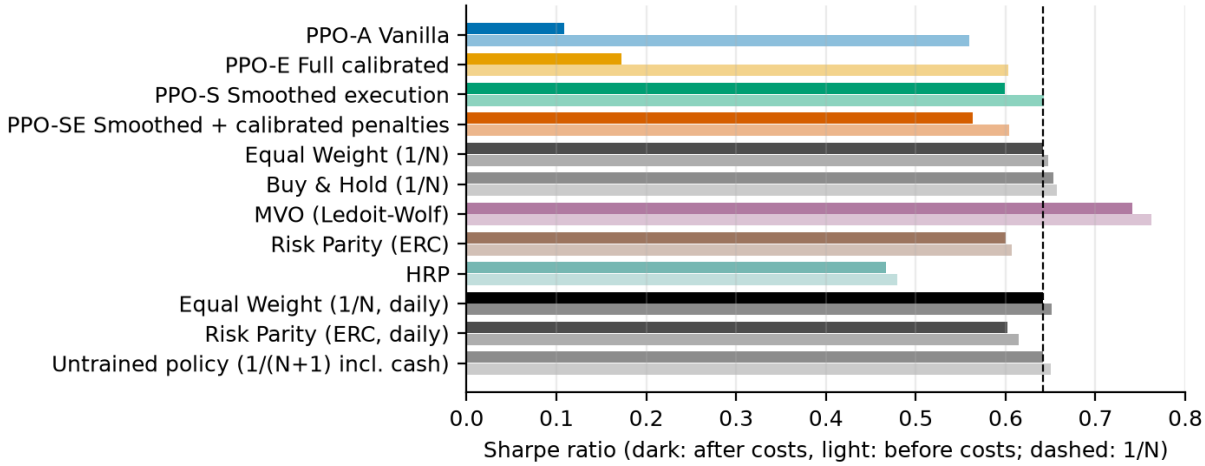


Figure 2: Sharpe ratio before (light) and after (dark) costs under next-open execution. The dashed line is net 1/N. No agent clears it even before costs.

6.5 Is any risk reduction learned?

Table 12: Is de-risking state-dependent? Average cash weight by regime, and the correlation between cash weight and the portfolio's trailing realised volatility (mean ± std over seeds).

Variant	Cash weight, calm (%)	Cash weight, stress (%)	Corr(cash, trailing 21d vol)
PPO-A · return only (λ_1=0, λ_2=0)	7.4 ± 0.9	7.0 ± 2.2	−0.09 ± 0.06
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	10.3 ± 3.8	10.7 ± 2.9	−0.01 ± 0.09
PPO-S · smoothed execution (λ_1=0, λ_2=0)	10.6 ± 1.5	10.4 ± 2.7	−0.08 ± 0.01
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	15.0 ± 1.2	13.9 ± 2.7	−0.11 ± 0.05

No agent de-risks when risk rises (Table 12). Every variant holds about the same cash on stress days as on calm days (within 1.1 points), and the correlation between an agent's cash weight and its own trailing volatility is between −0.11 and −0.01 on average, the wrong sign for de-risking.

Table 13: Exposure-matched control. Each agent (seed-averaged returns) against 1/N scaled to the agent's average invested fraction, with the remainder in zero-yield cash.

Variant	Avg cash (%)	Sharpe: agent vs control	MaxDD %: agent vs control	Stress $CVaR_{95}$ %: agent vs control	Worst stress day %: agent vs control
PPO-A · return only (λ_1=0, λ_2=0)	7.4	0.12 vs 0.64 (p=0.00)	26.3 vs 18.4	2.72 vs 2.38	−5.81 vs −5.20
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	10.3	0.18 vs 0.64 (p=0.00)	21.3 vs 17.8	2.35 vs 2.31	−4.85 vs −5.04
PPO-S · smoothed execution (λ_1=0, λ_2=0)	10.6	0.61 vs 0.64 (p=0.62)	19.0 vs 17.8	2.49 vs 2.30	−5.12 vs −5.02
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	14.9	0.58 vs 0.64 (p=0.28)	18.9 vs 17.0	2.19 vs 2.19	−4.27 vs −4.78

The exposure-matched control makes the point concrete (Table 13). 1/N scaled to SE's average invested fraction (85%) has a lower maximum drawdown than SE (17.0% against 18.9% on seed-averaged returns) and a Sharpe ratio that is not distinguishable from it; the same holds for S. On the 391 stress days, SE's CVaR is no lower than the control's (2.20% against 2.19%), though its worst stress day is milder (−4.3% against −4.8%).

Table 14: Regime-conditional statistics of net daily returns (3,136 calm and 391 stress days out of sample). PPO rows average the per-seed statistics. *Note:* Regime labels are never observed by any strategy; they are used only for this analysis.

Strategy	Calm: ann. mean (%)	Calm: ann. vol (%)	Calm: $CVaR_{95}$ (%)	Calm: worst day (%)	Stress: ann. mean (%)	Stress: ann. vol (%)	Stress: $CVaR_{95}$ (%)	Stress: worst day (%)
PPO-A · return only (λ_1=0, λ_2=0)	6.6	8.57	1.18	−3.28	−43.3	16.42	2.97	−7.03
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	6.4	7.42	1.05	−2.97	−38.7	13.38	2.48	−5.57
PPO-S · smoothed execution (λ_1=0, λ_2=0)	10.5	7.55	1.04	−2.83	−38.0	13.76	2.53	−5.18
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	9.0	6.78	0.94	−2.67	−33.7	12.32	2.24	−4.49
Equal Weight (1/N)	11.7	7.74	1.06	−2.80	−42.8	14.24	2.57	−5.62
Buy & Hold (1/N)	11.7	7.76	1.06	−2.74	−42.4	14.16	2.57	−5.49
MVO (Ledoit-Wolf)	13.8	12.33	1.88	−8.66	−19.4	21.24	3.61	−6.69
Risk Parity (ERC)	8.8	6.23	0.86	−1.98	−32.7	11.43	2.06	−4.27
HRP	7.2	5.76	0.79	−1.57	−29.6	10.97	1.97	−4.51
Equal Weight (1/N, daily)	11.8	7.80	1.06	−2.81	−43.4	14.55	2.63	−5.83
Risk Parity (ERC, daily)	8.8	6.20	0.86	−2.02	−32.5	11.38	2.04	−4.23
Untrained policy (1/(N+1) incl. cash)	10.8	7.09	0.97	−2.55	−39.5	13.23	2.39	−5.31

Every strategy loses money on the stress days (Table 14), from −19% annualised for mean-variance to −43% for 1/N. The smoothed agents lose somewhat less than 1/N (S −38%, SE −34%), consistent with their cash buffers, and risk parity and HRP have the lowest stress-period tail risk. In the 2022 fold, when stocks and bonds fell together, every agent did about as badly as or worse than 1/N (Sharpe −1.40 to −1.66 against −1.38; appendix Table 21).

6.6 The untrained policy

The untrained policy matches 1/N (Sharpe 0.64), has the smallest maximum drawdown of all strategies (18.4%) and is at least as good as every trained agent: SE − untrained is −0.08 ([−0.23, 0.06]) and A − untrained −0.53 ([−0.79, −0.31]). Its cash weight (9%) is close to S's (11%), and the timing decomposition shows that what the smoothed agents add to that starting point, in tilts and timing, did not pay. Training moves the policy away from a diversified, partly de-levered prior mainly by adding turnover; smoothing and penalties help mostly by limiting how far it moves. The untrained policy also matches 1/N to within 0.01 Sharpe in all four independent simulated markets of §7.3, where PPO-S is never significantly different from 1/N.

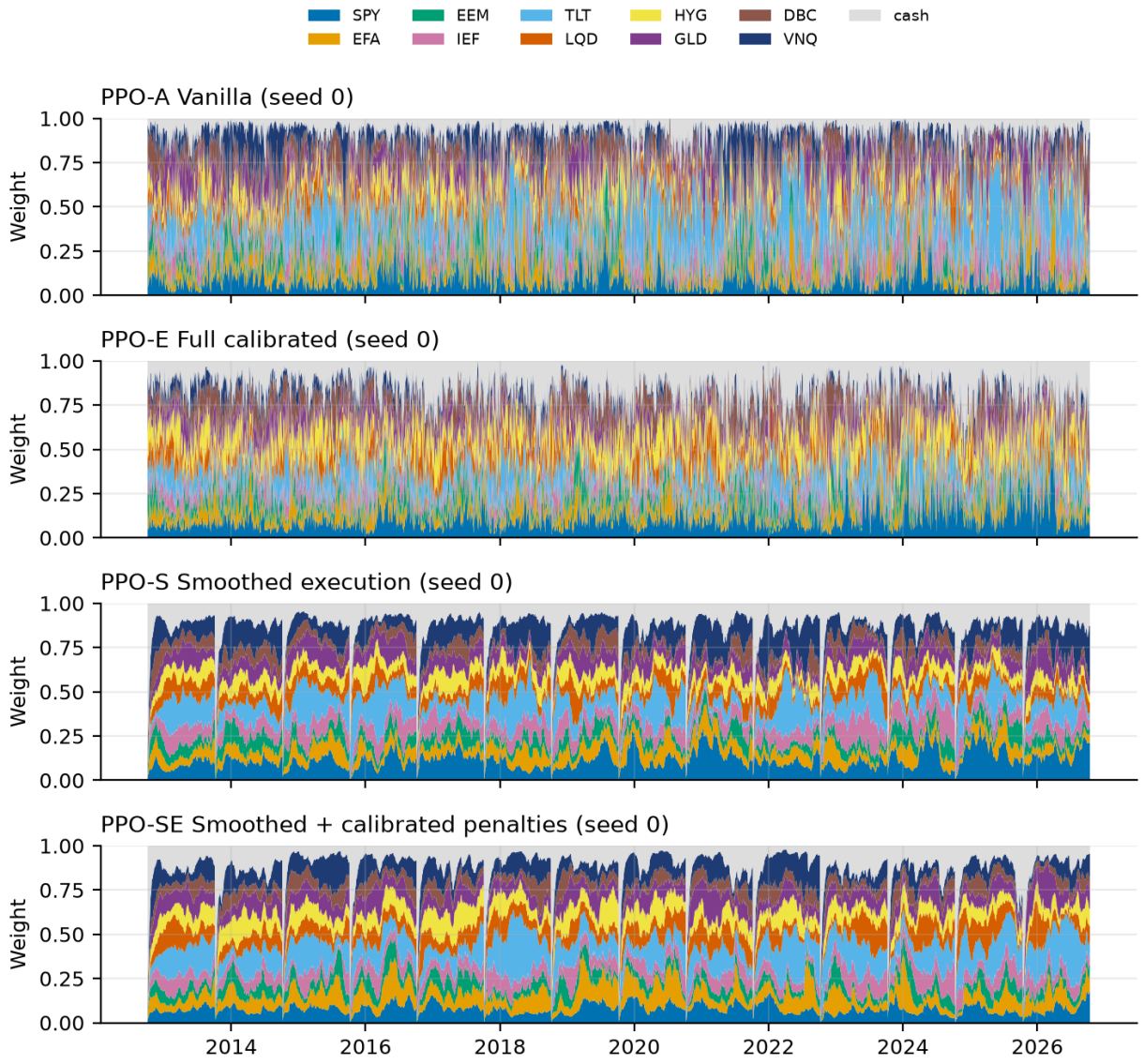


Figure 3: Daily allocations of PPO-A, E, S and SE (seed 0). The unsmoothed agents re-allocate a large part of the book every day; the smoothed agents move slowly. The wedges at fold boundaries are the ramp from cash at the start of each test fold.

7 Robustness

7.1 Costs, impact, fill delay and the starting book

Table 15: Evaluation-only sensitivity: net Sharpe ratio of the same trained models (and of every baseline) re-run on the out-of-sample windows under perturbed execution assumptions. PPO: mean over 5 seeds. *Note:* Costs ×k scales commission, half-spread and both impact coefficients; "No costs" also lifts the participation cap. "No impact" keeps fees and spread only. "Fill next close" adds a further session of delay. *"Same close" is the non-tradable legacy convention, shown only to measure how much it flatters the same policies. "Start from 1/N": every fold inherits a 1/N book instead of starting in cash. Policies were trained under the base assumptions.

Strategy	Base	No costs	Costs ×0.5	Costs ×2	Costs ×3	No impact	Impact ×3	Fill next close	Same close*	Start from 1/N
PPO-A · return only (λ_1=0, λ_2=0)	0.11	0.56	0.33	−0.34	−0.79	0.18	−0.04	0.12	0.26	0.12
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.17	0.60	0.39	−0.26	−0.68	0.23	0.05	0.12	0.20	0.18
PPO-S · smoothed execution (λ_1=0, λ_2=0)	0.60	0.64	0.62	0.56	0.51	0.60	0.59	0.59	0.61	0.62
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0.56	0.60	0.58	0.52	0.48	0.57	0.56	0.56	0.58	0.58
Equal Weight (1/N)	0.64	0.65	0.64	0.64	0.63	0.64	0.64	0.64	0.65	0.65
Buy & Hold (1/N)	0.65	0.67	0.65	0.65	0.64	0.65	0.65	0.65	0.66	0.67
MVO (Ledoit-Wolf)	0.74	0.74	0.75	0.72	0.70	0.75	0.73	0.73	0.75	0.74
Risk Parity (ERC)	0.60	0.61	0.60	0.59	0.59	0.60	0.60	0.60	0.60	0.61
HRP	0.47	0.48	0.47	0.45	0.44	0.47	0.46	0.47	0.46	0.48
Equal Weight (1/N, daily)	0.64	0.65	0.65	0.63	0.62	0.64	0.64	0.64	0.65	0.65
Risk Parity (ERC, daily)	0.60	0.61	0.61	0.59	0.58	0.60	0.60	0.60	0.61	0.61
Untrained policy (1/(N+1) incl. cash)	0.64	0.65	0.65	0.63	0.62	0.64	0.64	0.64	0.66	0.65

Table 16: Do the comparisons survive the perturbations? Sharpe differences with 95% two-level bootstrap intervals (1,000 replications) under each scenario of the previous table. *Note:* * $p < 0.05$, ** $p < 0.01$, uncorrected (these are robustness checks, not new primary tests).

Scenario	P1: PPO-E − PPO-A	P2: PPO-S − PPO-A	P3: PPO-SE − 1/N	P4: PPO-A − 1/N	S-1/N: PPO-S − 1/N	SE-untrained: PPO-SE − Untrained policy
Base	+0.06 [−0.11, 0.25]	+0.49 [0.26, 0.72]**	−0.08 [−0.23, 0.05]	−0.53 [−0.77, −0.32]**	−0.04 [−0.20, 0.09]	−0.08 [−0.23, 0.05]
No costs	+0.04 [−0.15, 0.23]	+0.08 [−0.14, 0.29]	−0.04 [−0.20, 0.09]	−0.09 [−0.29, 0.12]	−0.01 [−0.17, 0.12]	−0.05 [−0.20, 0.09]
Costs ×0.5	+0.05 [−0.13, 0.24]	+0.29 [0.07, 0.50]*	−0.06 [−0.22, 0.07]	−0.31 [−0.53, −0.10]**	−0.02 [−0.18, 0.11]	−0.06 [−0.22, 0.07]
Costs ×2	+0.08 [−0.10, 0.28]	+0.90 [0.65, 1.16]**	−0.11 [−0.27, 0.02]	−0.98 [−1.25, −0.73]**	−0.08 [−0.24, 0.05]	−0.11 [−0.27, 0.02]
Costs ×3	+0.10 [−0.08, 0.31]	+1.30 [1.02, 1.61]**	−0.15 [−0.30, −0.02]*	−1.42 [−1.74, −1.12]**	−0.12 [−0.28, 0.01]	−0.14 [−0.30, −0.01]
No impact	+0.05 [−0.13, 0.24]	+0.42 [0.19, 0.64]**	−0.08 [−0.23, 0.06]	−0.46 [−0.70, −0.25]**	−0.04 [−0.20, 0.09]	−0.08 [−0.23, 0.06]
Impact ×3	+0.09 [−0.09, 0.28]	+0.63 [0.41, 0.86]**	−0.08 [−0.24, 0.05]	−0.68 [−0.93, −0.46]**	−0.05 [−0.21, 0.08]	−0.08 [−0.24, 0.05]
Fill next close	−0.01 [−0.20, 0.22]	+0.47 [0.24, 0.72]**	−0.08 [−0.24, 0.06]	−0.52 [−0.77, −0.29]**	−0.05 [−0.21, 0.08]	−0.08 [−0.24, 0.06]
Same close*	−0.06 [−0.25, 0.16]	+0.35 [0.11, 0.61]**	−0.07 [−0.23, 0.06]	−0.38 [−0.64, −0.14]**	−0.03 [−0.19, 0.10]	−0.08 [−0.23, 0.05]
Start from 1/N	+0.07 [−0.11, 0.26]	+0.50 [0.28, 0.73]**	−0.07 [−0.22, 0.06]	−0.54 [−0.78, −0.32]**	−0.04 [−0.20, 0.08]	−0.07 [−0.22, 0.06]

The base scenario reproduces the main results exactly, so differences in Table 15 come only from the perturbation. Three findings stand out.

R1. No agent beats 1/N even without costs. With every friction removed, the agents' Sharpe ratios are 0.56–0.64, against 0.65 for 1/N and the untrained policy: the same conclusion as the pre-specified P5, now for all four variants and without relying on the gross-up of costs.

R2. Smoothing helps only by saving costs. The benefit of smoothing (P2) grows with costs, from +0.08 ([−0.14, 0.29]) with no costs to +0.29 at half costs, +0.49 in the base case and +1.30 at triple costs. Without costs there is nothing for smoothing to save, and no timing skill for it to protect.

R3. The conclusions do not depend on the exact cost model. P1 (penalties) is never significant (estimates −0.06 to +0.10). P4 (vanilla PPO worse than 1/N) is significant in every scenario with costs. P3 (SE versus 1/N) is never significantly positive; its point estimate falls from −0.04 without costs to −0.15 at triple costs, where it becomes significantly negative. Removing impact entirely (fees and spread only) or tripling it changes the smoothed agents' Sharpe ratios by less than 0.01 and the unsmoothed agents' by up to 0.15. Filling one session later, at the next close, changes no conclusion. Starting every fold from an inherited 1/N book instead of cash raises the smoothed agents' Sharpe ratios by about 0.02, about as much as it raises 1/N's own (+0.01).

7.2 Warm-started versus from-scratch training

Table 17: Warm-start control. Selected folds retrained from scratch with the fold-0 budget on exactly the same rolling training window and PPO seed, compared fold by fold with the warm-started chain. Intervals bootstrap the (seed, fold) pairs.

Variant	Metric (per fold)	Warm-started chain	From scratch	Scratch − warm [95% CI]	Pairs where scratch is higher
PPO-S · smoothed execution (λ_1=0, λ_2=0)	Sharpe	1.34	1.22	−0.12 [−0.38, 0.19]	3/9
PPO-S · smoothed execution (λ_1=0, λ_2=0)	Ann. return (%)	9.13	8.05	−1.07 [−2.91, 1.26]	2/9
PPO-S · smoothed execution (λ_1=0, λ_2=0)	MaxDD (%)	4.1	4.1	+0.0 [−0.4, 0.4]	3/9
PPO-S · smoothed execution (λ_1=0, λ_2=0)	Turnover (%/day)	4.7	3.3	−1.4 [−1.7, −1.1]	0/9

Fine-tuning each fold from the previous fold's policy could carry stale structure forward or leave later folds under-trained. We retrained PPO-S from scratch with the full fold-0 budget (102,400 steps) on folds 4, 8 and 13 (test windows starting October 2016, 2020 and 2025) for seeds 0–2, on exactly the same training windows, and compared each fold with its warm-started counterpart (Table 17). The from-scratch models have a slightly lower per-fold Sharpe ratio (−0.12, [−0.38, 0.19]; higher in 3 of 9 pairs) and the same drawdown, so warm starting is not what keeps the trained agents from beating 1/N. They trade less (3.3% of the book a day against 4.7%; the interval excludes zero): turnover builds up along a warm-started chain, which fits the reading of §6.6 that training mainly adds turnover. This control covers one variant and three folds, not every chain.

7.3 Independent simulated markets

Table 18: Independent simulated markets under next-open execution (4 annual walk-forward folds each, warm-started chains as in the ETF study, 2 training seeds per variant). Intervals are two-level bootstraps (2,000 replications). *Note:* Standard markets use the generator of §4.2 with new data seeds. The bear market has negative calm-regime drift (−8% to 0% a year) and stress twice as often, so holding cash pays.

Market	1/N Sharpe	Untrained Sharpe	PPO-S Sharpe	PPO-SE Sharpe	S − 1/N [95% CI]	SE − S [95% CI]	MaxDD %: 1/N / S / SE	Cash % calm → stress: S; SE
Standard, data seed 101	−0.27	−0.27	−0.57 ± 0.17	−0.30 ± 0.06	−0.29 [−0.66, 0.07]	+0.27 [−0.06, 0.64]	51.6 / 52.0 / 36.9	19 → 33; 30 → 37
Standard, data seed 202	−0.69	−0.69	−0.43 ± 0.15	−0.79 ± 0.14	+0.25 [−0.11, 0.62]	−0.36 [−0.77, 0.01]	70.2 / 48.5 / 60.5	28 → 32; 21 → 26
Standard, data seed 303	−0.56	−0.55	−0.66 ± 0.09	−0.61 ± 0.09	−0.10 [−0.36, 0.12]	+0.05 [−0.22, 0.38]	57.9 / 60.4 / 40.2	19 → 15; 44 → 43
Bear, data seed 404	−0.01	−0.01	0.09 ± 0.06	0.13 ± 0.08	+0.10 [−0.17, 0.38]	+0.04 [−0.22, 0.33]	31.3 / 19.0 / 14.1	29 → 28; 47 → 40

The simulated market of earlier versions was a single draw from the generator. We drew three new standard markets and a bear market, in which calm-regime drift is negative and stress occurs twice as often, so that an agent that holds cash in bad times has something to gain (Table 18). The runs are small (four folds, two seeds) and show whether the ETF pattern is an accident of one data set; they do not estimate effects precisely. Four results hold in every market. The untrained policy matches 1/N to within 0.01 Sharpe. PPO-S is never significantly different from 1/N (estimates −0.29 to +0.25). The effect of the tail-risk penalties, SE − S, changes sign across markets (−0.36 to +0.27) and is never significant. PPO-SE has a smaller maximum drawdown than 1/N in all four markets (bear market: 14% against 31%), in line with its higher average cash weight, as in the ETF study. Its cash response to stress is not consistent: against the ground-truth regime labels, PPO-S and PPO-SE each raise cash on stress days in two standard markets (by about 5–13 points), but in the bear market, where de-risking pays most, neither does (S 29% → 28%, SE 47% → 40%). Where holding cash pays, the trained agents gain from holding more of it on average, not from timing it.

Table 19: How often the reward clip (±10 after scaling by 100) binds. Training-window replay: each fold's final policy run deterministically over its own training window in consecutive 252-day episodes (the training episode length), all seeds.

Variant	Clipped steps, training-window replay	Range of unclipped reward (training replay)	Clipped steps, out of sample
PPO-A · return only (λ_1=0, λ_2=0)	3 / 70,490	[−10.6, 10.1]	0 / 17,635
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	5 / 70,490	[−11.7, 7.4]	0 / 17,635
PPO-S · smoothed execution (λ_1=0, λ_2=0)	0 / 70,490	[−7.5, 7.1]	0 / 17,635
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0 / 70,490	[−8.7, 5.5]	0 / 17,635

7.4 Reward clipping

The reward clip almost never binds (Table 19): never out of sample, and in at most 5 of about 70,000 training-window steps per variant (all in the unsmoothed variants, on days of extreme losses). Clipping cannot explain any of the results.

8 What remains of the earlier results

Versions 1–5 also ran variants that were not rerun: CVaR-only and drawdown-only penalties at $\lambda = 1$ (B, C, D), a turnover penalty (T), RCPO with loose and with 1/N-matched targets (R, RT, SRT), a volatility-targeting overlay, and a controlled 2×2 ablation on a simulated market with validation-based checkpoint selection. All of these used the same-close convention. Their absolute numbers are therefore not tradable and are not used for any claim here. Comparisons among slowly trading strategies, which the convention barely affects (§5.3), and qualitative patterns that the corrected rerun reproduces remain informative. Three such patterns are worth recording:

- **Training past a short budget made agents worse out of sample.** On the legacy simulated market, vanilla PPO's Sharpe ratio fell from 1.24 at 50k steps to 1.04 at 150k while its in-sample Sharpe kept rising, and the untrained policy beat every trained agent at every budget (appendix Table 26). The corrected ETF study again finds the untrained policy unbeaten (§6.6).
- **Validation-based checkpoint selection interacted with the penalties**, favouring early, less-trained checkpoints for penalised variants. This is why the ETF study uses no selection.
- **Binding RCPO constraints did not produce state-dependent de-risking** (appendix Table 24), and neither did any fixed penalty, the same conclusion as §6.5.

The legacy paper, its tables and its outputs are preserved in the repository (`paper/legacy/`, `outputs_real/`, `outputs_real_followup/`, `outputs/`) with their configurations pinned to the same-close convention, so every earlier number remains reproducible.

9 Discussion

9.1 Fill conventions matter most for the agents that look best

The same-close convention changed the verdict on exactly the strategies that trade most. It turned a gross deficit into a gross advantage for every unsmoothed agent and produced a timing story that a static-tilt decomposition and a paired bootstrap appeared to confirm. Neither check could catch it, because both take the simulated fills as given. Two practices would have caught it earlier: specifying the fill price relative to the information set in the environment's interface (here, separate decision and execution prices, with a test that perturbs everything after the decision), and routinely

evaluating the same trained policies under a delayed fill (§7.1). We suspect the problem is common in applied DRL allocation, where daily close-to-close environments are the default.

9.2 Execution design versus reward shaping

Under tradable execution, the agents' problem is not that they find a signal and trade it away (the earlier story) but that they find no tradable signal and still trade heavily. Exploration noise in training keeps turnover high, the policy re-targets every day, and every change of view is paid for. Slowing execution is the only intervention that moved the Sharpe ratio significantly, and it did so by limiting damage rather than by enabling skill. Tail-risk penalties act on a different margin: they lower turnover a little and hold more cash, which lowers CVaR and drawdown about as much as the same cash in 1/N does.

9.3 Penalties as regularisers towards the prior

PPO's softmax policy starts as a diversified, partly de-levered portfolio, and that portfolio is hard to beat. Return-only training pulls the policy towards concentrated, high-turnover allocations that fit the training window; drawdown penalties and smoothing make those excursions costly or slow, so the policy stays nearer its prior. That reading explains why the penalties reduce risk without improving the Sharpe ratio, why no agent de-risks with volatility, and why the untrained policy is the strongest "agent" in both the corrected and the legacy studies.

9.4 What a better design would need

An allocator that adds value over 1/N under tradable execution would need a signal that survives the overnight gap and costs; a trading speed matched to that signal's horizon, learned (as in an aim-portfolio formulation [46]) rather than fixed; and risk information it can act on before losses arrive, for instance forecast volatility in the state and a critic that estimates the CVaR of the return-to-go [20, 21, 54] rather than a penalty on realised history. It would also need regularisation against fitting the training window, since performance fell with training in every setting we examined.

10 Limitations

- **Not an untouched hold-out.** The 2012–2026 test period informed design choices in versions 1–5. The analysis plan fixes the analysis of the rerun, not the history of the design.
- **One market-data universe.** Ten ETFs at daily frequency, chosen in 2026 from funds that still exist (survivorship and selection). Conclusions may differ for single stocks, futures, intraday data or other periods.
- **Power.** With 5 seeds and 3,527 days, the primary tests can detect Sharpe differences of about 0.21–0.34 with 80% power. Smaller effects of the penalties, or a small advantage or disadvantage of SE against 1/N, cannot be ruled out.
- **Costs and liquidity.** Cost parameters are plausible for institutional ETF trading, not calibrated to a desk. ADV is computed from back-adjusted prices, which understates early-sample dollar volume and so overstates early impact costs; this is conservative and mostly penalises high-turnover strategies. Fixed-share orders can leave small negative cash after gaps, financed at 0%.
- **Design choices not tuned.** PPO hyper-parameters, training budgets, penalty multipliers and the smoothing speed are fixed, not tuned per variant; warm-started chains accumulate

training across folds (§7.2 checks a subset). Each test fold starts in cash, so smoothed agents spend about a month per fold ramping up (§7.1 tests an inherited book).

- **Simulated markets.** Two seeds per variant and four folds per market; the generator is ours, and its latent alpha may be easier or harder to learn than real signals.
- **No distributional or risk-sensitive RL baseline.** We test reward penalties, not methods that optimise a risk measure of the return-to-go [21, 54].
- **Stress labels** are named historical episodes dated with hindsight; they are used only for analysis.

11 Conclusion

We evaluated tail-risk reward penalties for PPO portfolio allocators under tradable next-open execution, realistic frictions, walk-forward retraining, five training seeds and pre-specified inference. The penalties reduce tail risk relative to unpenalised PPO, but only as much as holding the same cash in 1/N would; they do not detectably improve the Sharpe ratio, and no agent learns to reduce risk when risk rises. Slowing execution is the only design change with a significant effect, and the best agent is then neither better than nor shown to be equivalent to equal weighting. A pre-cost timing edge reported in earlier versions of this study was an artifact of filling at a close the agent had already seen, and is withdrawn. For DRL allocators, the fill convention and the trading process deserve at least as much scrutiny as the reward, and the untrained network's own prior is a benchmark worth reporting.

12 Reproducibility

All code, configuration and outputs are in the public repository `https://github.com/NaMEleSs0616/risk-constrained-drl-allocation`, release `v6.0`. The release includes the analysis plan, a pinned dependency lock file (`requirements-lock.txt`), the SHA-256 hash of the downloaded price file, and a manifest of every committed output with its hash (`MANIFEST.json`). The environment follows the Gymnasium API [42]; the test suite (158 tests) covers cost accounting, execution timing and look-ahead, CVaR and drawdown estimators, baselines, RCPO updates, metrics and inference. The corrected ETF study (20 chains, 9.5M environment steps) took 10.1 hours on four CPU worker processes on an Apple M3 laptop.

```
pip install -r requirements-lock.txt && pytest
python scripts/download_data.py --start 2006-01-01 --end 2026-10-15          # ETF data (Yahoo Finance)
python scripts/run_real_experiments.py --config config/real_etf_nextopen.yaml # -> outputs_real_nextopen/
python scripts/inference_report.py --results outputs_real_nextopen            # pre-specified tests
python scripts/extra_analysis.py --results outputs_real_nextopen
python scripts/sensitivity_analysis.py --results outputs_real_nextopen        # evaluation-only robustness
scripts/run_robustness.sh                                                     # simulated markets + warm-start control
python scripts/synthetic_robustness_summary.py --root outputs_syn_nextopen
python scripts/make_figures.py --results outputs_real_nextopen --out paper/figures/nextopen \
    --training-folds fold_00 --alloc PPO-A PPO-E PPO-S PPO-SE
python scripts/build_paper.py --results no=outputs_real_nextopen sm=outputs_syn_nextopen \
    real=outputs_real fu=outputs_real_followup syn=outputs
python scripts/build_latex.py && (cd paper/latex && tectonic main.tex)       # PDF
```

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Appendix

.1 A. Corrected ETF study

Table 20: ETF study: walk-forward fold calendar and the share of named-stress days in each window.

Fold	Training window	Stress, train	Test window	Stress, test
F0	2008-09-25 → 2012-09-24	21.2%	2012-10-02 → 2013-10-03	0.0%
F1	2009-09-25 → 2013-09-26	10.0%	2013-10-04 → 2014-10-03	0.0%
F2	2010-09-27 → 2014-09-26	5.1%	2014-10-06 → 2015-10-05	11.9%
F3	2011-09-26 → 2015-09-28	3.5%	2015-10-06 → 2016-10-04	12.3%
F4	2012-09-25 → 2016-09-27	6.1%	2016-10-05 → 2017-10-04	0.0%
F5	2013-09-27 → 2017-09-27	6.1%	2017-10-05 → 2018-10-04	7.9%
F6	2014-09-29 → 2018-09-27	7.5%	2018-10-05 → 2019-10-07	21.8%
F7	2015-09-29 → 2019-09-30	10.6%	2019-10-08 → 2020-10-06	9.5%
F8	2016-09-28 → 2020-09-29	9.8%	2020-10-07 → 2021-10-06	0.0%
F9	2017-09-28 → 2021-09-29	9.8%	2021-10-07 → 2022-10-06	76.2%
F10	2018-09-28 → 2022-09-29	26.9%	2022-10-07 → 2023-10-09	1.6%
F11	2019-10-01 → 2023-10-02	21.8%	2023-10-10 → 2024-10-09	0.0%
F12	2020-09-30 → 2024-10-02	19.4%	2024-10-10 → 2025-10-13	13.9%
F13	2021-09-30 → 2025-10-06	22.9%	2025-10-14 → 2026-10-14	0.0%

Table 21: Out-of-sample Sharpe ratio in each walk-forward fold (PPO: mean over seeds).

Strategy	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13
PPO-A · return only (λ_1=0, λ_2=0)	−0.84	0.35	−0.67	0.26	1.02	−0.49	0.43	0.62	0.87	−1.66	0.07	1.30	0.80	0.07
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	−0.62	0.34	−1.06	0.66	0.65	−0.42	0.66	0.56	1.08	−1.66	0.16	1.52	0.74	0.29
PPO-S · smoothed execution (λ_1=0, λ_2=0)	−0.28	0.81	−0.34	0.76	1.62	−0.15	1.80	0.67	1.90	−1.40	0.68	2.34	1.25	0.64
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	−0.33	0.87	−0.46	0.86	1.39	−0.13	1.84	0.72	1.64	−1.51	0.67	2.35	1.24	0.67
Equal Weight (1/N)	−0.04	1.02	−0.42	0.99	1.39	0.16	1.53	0.57	1.68	−1.38	0.62	2.23	1.34	1.01
Buy & Hold (1/N)	0.03	1.02	−0.39	0.93	1.40	0.17	1.63	0.53	1.75	−1.36	0.61	2.20	1.38	0.94
MVO (Ledoit-Wolf)	0.97	1.49	0.24	−0.05	0.60	0.65	0.72	0.12	0.49	0.14	−0.31	2.46	2.39	1.05
Risk Parity (ERC)	−0.29	1.24	−0.41	1.22	1.03	0.14	2.66	0.55	0.88	−1.60	0.55	2.24	1.30	1.07
HRP	−0.62	1.40	−0.37	1.32	0.72	−0.26	3.24	0.45	0.22	−1.86	0.67	2.43	1.25	0.37
Equal Weight (1/N, daily)	−0.04	1.01	−0.40	0.99	1.38	0.17	1.53	0.57	1.69	−1.38	0.62	2.22	1.35	1.04
Risk Parity (ERC, daily)	−0.29	1.22	−0.38	1.25	1.04	0.13	2.67	0.57	0.88	−1.62	0.55	2.22	1.30	1.11
Untrained policy (1/(N+1) incl. cash)	−0.04	1.01	−0.40	0.99	1.38	0.17	1.53	0.56	1.69	−1.39	0.62	2.22	1.34	1.04

.2 B. Legacy results under the same-close convention (not tradable)

The tables in this section reproduce versions 1–5. Every order was filled at the close of the day whose data the policy had observed, so the absolute numbers, above all those of fast-trading agents,

Table 22: Every PPO training seed (stitched out-of-sample track).

Variant	Seed	Cum. ret (%)	Sharpe	Sortino	MaxDD (%)	Calmar	$CVaR_{95}$ (%)	Turnover (%/day)	Worst day (%)
PPO-A · return only (λ_1=0, λ_2=0)	0	14.7	0.15	0.21	29.2	0.03	1.41	47.0	−7.86
PPO-A · return only (λ_1=0, λ_2=0)	3	4.7	0.08	0.11	28.0	0.01	1.36	50.6	−5.88
PPO-A · return only (λ_1=0, λ_2=0)	1	8.0	0.11	0.14	26.3	0.02	1.54	48.6	−7.62
PPO-A · return only (λ_1=0, λ_2=0)	2	13.6	0.14	0.20	27.5	0.03	1.39	48.3	−6.99
PPO-A · return only (λ_1=0, λ_2=0)	4	1.5	0.06	0.09	30.0	0.00	1.62	48.1	−6.81
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	2	15.8	0.18	0.24	22.0	0.05	1.12	39.2	−4.54
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	1	11.1	0.13	0.17	21.1	0.04	1.39	42.2	−7.34
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0	24.1	0.24	0.33	20.6	0.08	1.17	37.6	−4.11
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	3	6.3	0.09	0.13	23.1	0.02	1.32	41.8	−6.27
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	4	24.5	0.23	0.31	23.8	0.07	1.30	41.4	−5.59
PPO-S · smoothed execution (λ_1=0, λ_2=0)	1	56.4	0.43	0.59	20.7	0.16	1.25	4.6	−4.91
PPO-S · smoothed execution (λ_1=0, λ_2=0)	0	101.7	0.67	0.93	20.1	0.26	1.19	4.3	−4.42
PPO-S · smoothed execution (λ_1=0, λ_2=0)	2	102.9	0.62	0.87	17.9	0.29	1.28	4.5	−5.16
PPO-S · smoothed execution (λ_1=0, λ_2=0)	3	100.6	0.62	0.86	19.1	0.27	1.29	4.6	−5.65
PPO-S · smoothed execution (λ_1=0, λ_2=0)	4	114.7	0.66	0.91	19.6	0.29	1.31	4.5	−5.76
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	0	55.9	0.45	0.61	21.8	0.15	1.18	3.8	−4.44
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	1	62.3	0.49	0.67	18.9	0.19	1.14	3.9	−5.08
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	2	84.9	0.62	0.86	17.1	0.26	1.14	4.0	−3.92
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	3	81.8	0.62	0.86	18.7	0.23	1.11	4.0	−3.86
PPO-SE · smoothed + calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	4	94.0	0.64	0.89	18.8	0.26	1.15	3.7	−5.15

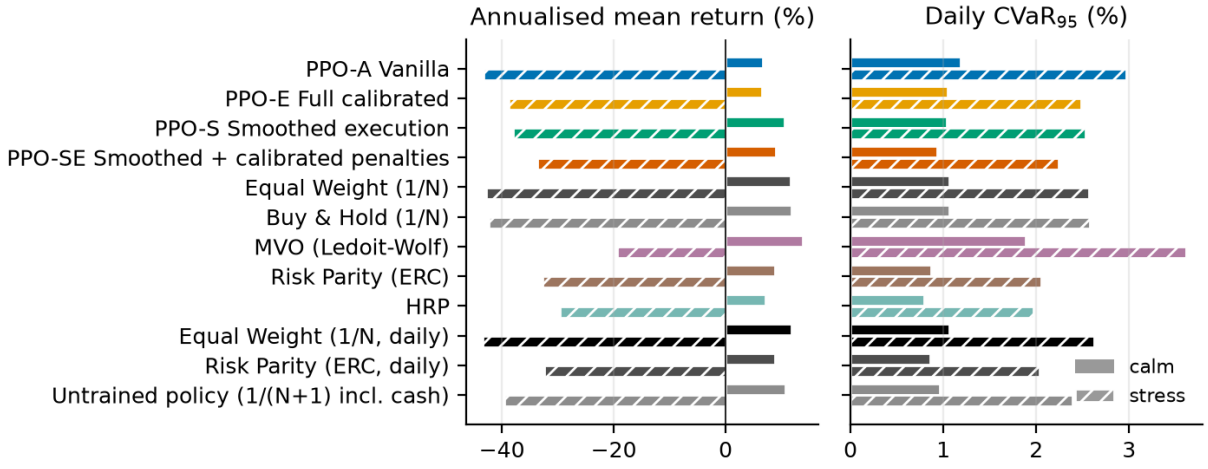


Figure 4: Regime-conditional annualised mean return and daily $CVaR_{95}$ under next-open execution (solid: calm; hatched: named stress episodes).

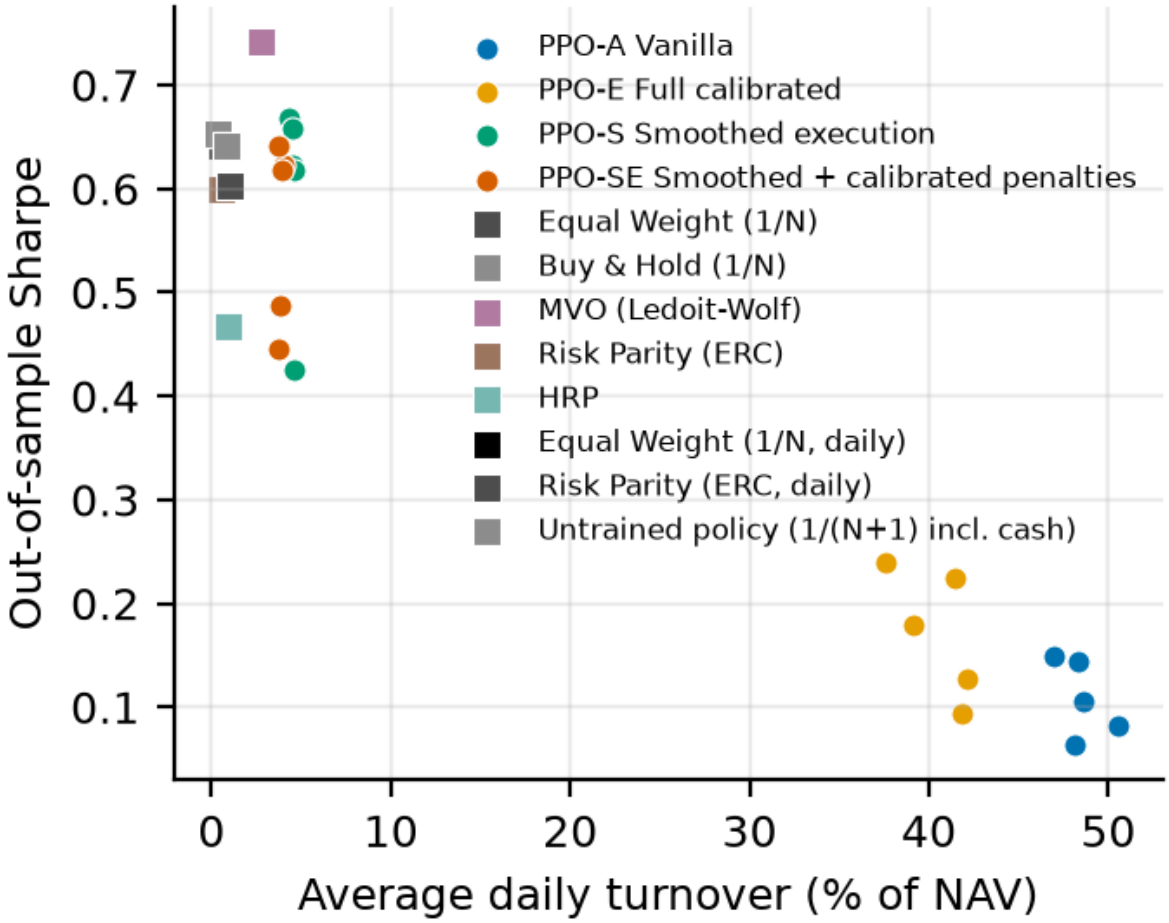


Figure 5: Turnover against out-of-sample Sharpe for each PPO seed (circles) and baseline (squares), next-open execution.

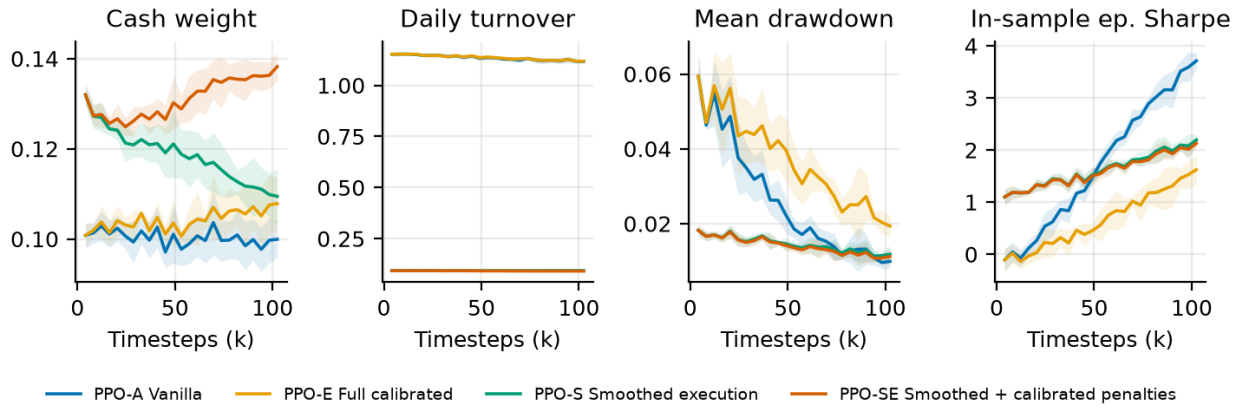


Figure 6: Fold-0 training dynamics under next-open execution (from scratch, 102k steps; mean ± std over seeds).

are not achievable. They are kept for transparency and for the qualitative patterns discussed in §8.

Table 23: Out-of-sample performance on the stitched walk-forward test period (14 folds, 3,527 trading days, net of all frictions). PPO rows: mean ± standard deviation across 3 training seeds. Bold marks the best value in each column. *Note:* $CVaR_{95}$ is the daily historical expected shortfall of net returns. Annualised return is geometric; Sharpe and Sortino use a zero risk-free rate.

Strategy	Cum. return (%)	Ann. return (%)	Ann. vol (%)	Sharpe	Sortino	MaxDD (%)	Calmar	$CVaR_{95}$ (%)
PPO-A · return only (λ_1=0, λ_2=0)	60.9 ± 13.3	3.4 ± 0.6	10.3 ± 0.7	0.38 ± 0.05	0.53 ± 0.08	22.2 ± 2.7	0.15 ± 0.02	1.51 ± 0.10
PPO-A · return only + vol-target overlay (λ_1=0, λ_2=0)	23.1 ± 13.4	1.5 ± 0.8	8.6 ± 0.3	0.21 ± 0.09	0.29 ± 0.13	22.0 ± 1.3	0.07 ± 0.04	1.31 ± 0.04
PPO-D · CVaR + drawdown (λ_1=1, λ_2=1)	40.8 ± 11.2	2.5 ± 0.6	8.0 ± 0.8	0.34 ± 0.05	0.47 ± 0.07	19.9 ± 1.0	0.12 ± 0.03	1.18 ± 0.11
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	49.4 ± 16.1	2.9 ± 0.8	8.7 ± 1.2	0.37 ± 0.07	0.51 ± 0.09	19.1 ± 2.3	0.15 ± 0.03	1.28 ± 0.17
PPO-T · turnover penalty (λ_1=0, λ_2=0)	53.8 ± 13.6	3.1 ± 0.6	10.2 ± 0.5	0.35 ± 0.06	0.49 ± 0.09	20.2 ± 1.2	0.15 ± 0.02	1.47 ± 0.09
PPO-R · RCPO, adaptive λ	44.1 ± 19.6	2.6 ± 1.0	9.3 ± 1.2	0.33 ± 0.13	0.46 ± 0.18	23.9 ± 2.6	0.11 ± 0.05	1.37 ± 0.17
Equal Weight (1/N)	109.7	5.4	8.8	0.65	0.89	19.9	0.27	1.30
Buy & Hold (1/N)	112.5	5.5	8.8	0.66	0.91	19.8	0.28	1.30
MVO (Ledoit-Wolf)	**265.8**	**9.7**	13.5	**0.75**	**1.03**	23.7	**0.41**	2.12
Risk Parity (ERC)	74.9	4.1	7.0	0.60	0.84	19.8	0.21	1.06
HRP	48.7	2.9	**6.6**	0.46	0.64	20.5	0.14	**0.98**
Untrained policy (1/(N+1) incl. cash)	100.3	5.1	8.1	0.66	0.91	**18.1**	0.28	1.19

Table 24: RCPO multipliers at the end of each fold's training.

Multiplier (mean over seeds)	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13
RT: λ_1 (CVaR)	0.00	0.00	0.05	0.13	0.32	0.58	0.67	0.78	0.61	0.67	0.67	0.53	0.59	0.55
RT: λ_2 (drawdown)	0.00	0.24	4.74	5.00	5.00	5.00	5.00	5.00	5.00	5.00	4.81	5.00	5.00	5.00
SRT: λ_1 (CVaR)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.02	0.00	0.00	0.00
SRT: λ_2 (drawdown)	0.00	0.00	0.10	0.01	0.63	1.34	1.53	1.29	0.89	0.93	0.56	0.82	1.99	2.44

Table 25: Out-of-sample performance on the stitched walk-forward test period (3 folds, 755 trading days, net of all frictions). PPO rows: mean ± standard deviation across 3 training seeds. Bold marks the best value in each column. *Note:* $CVaR_{95}$ is the daily historical expected shortfall of net returns. Annualised return is geometric; Sharpe and Sortino use a zero risk-free rate.

Strategy	Cum. return (%)	Ann. return (%)	Ann. vol (%)	Sharpe	Sortino	MaxDD (%)	Calmar	$CVaR_{95}$ (%)
PPO-A · return only (λ_1=0, λ_2=0)	128.0 ± 50.0	31.2 ± 9.5	26.6 ± 0.5	1.14 ± 0.26	1.84 ± 0.44	23.9 ± 1.5	1.30 ± 0.32	3.64 ± 0.20
PPO-B · CVaR only (λ_1=1, λ_2=0)	62.3 ± 19.0	17.4 ± 4.6	22.8 ± 0.8	0.82 ± 0.17	1.22 ± 0.28	20.3 ± 1.5	0.87 ± 0.27	3.29 ± 0.16
PPO-C · drawdown only (λ_1=0, λ_2=1)	113.0 ± 20.7	28.6 ± 4.1	22.4 ± 0.3	1.23 ± 0.13	1.96 ± 0.25	**17.7 ± 0.9**	1.62 ± 0.32	3.07 ± 0.10
PPO-D · CVaR + drawdown (λ_1=1, λ_2=1)	111.7 ± 13.2	28.4 ± 2.6	22.0 ± 1.0	1.25 ± 0.14	1.96 ± 0.23	18.3 ± 1.6	1.56 ± 0.24	3.07 ± 0.16
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	109.3 ± 15.7	27.9 ± 3.2	22.4 ± 0.3	1.21 ± 0.10	1.91 ± 0.19	18.4 ± 1.2	1.53 ± 0.28	3.10 ± 0.07
Equal Weight (1/N)	172.2	39.7	26.6	1.39	2.17	20.9	1.90	3.77
Buy & Hold (1/N)	**180.8**	**41.1**	27.4	**1.39**	**2.17**	21.6	**1.90**	3.88
MVO (Ledoit-Wolf)	132.6	32.5	27.3	1.17	1.80	27.3	1.19	3.72
Risk Parity (ERC)	131.6	32.4	23.1	1.33	2.06	17.8	1.82	3.29
HRP	88.5	23.6	**19.5**	1.18	1.82	17.9	1.32	**2.76**

.3 C. Environment parameters

Table 26: Selection-free robustness check (controlled study). Every agent evaluated out of sample at fixed training budgets, versus the untrained policy. Mean ± std over seeds.

Strategy	Checkpoint	Sharpe	MaxDD (%)	$CVaR_{95}$ (%)	Turnover (%/day)	Total cost (bp)
Initial policy (1/6 incl. cash)	untrained	1.38	17.8	3.12	1.2	36
PPO-A · return only (λ_1=0, λ_2=0)	50k	1.24 ± 0.24	22.2 ± 0.8	3.42 ± 0.09	37.6 ± 1.1	1539 ± 44
	100k	1.12 ± 0.22	24.7 ± 2.8	3.81 ± 0.06	57.7 ± 2.6	2774 ± 170
	150k (final)	1.04 ± 0.27	26.9 ± 2.0	4.04 ± 0.09	64.9 ± 4.1	3266 ± 244
	validation-selected	1.14 ± 0.26	23.9 ± 1.5	3.64 ± 0.20	40.2 ± 9.7	1859 ± 447
PPO-B · CVaR only (λ_1=1, λ_2=0)	50k	0.93 ± 0.16	18.9 ± 2.1	3.23 ± 0.24	29.4 ± 0.8	1073 ± 25
	100k	0.71 ± 0.12	21.5 ± 2.3	3.31 ± 0.36	42.2 ± 1.6	1692 ± 63
	150k (final)	0.66 ± 0.15	22.2 ± 0.9	3.32 ± 0.24	47.8 ± 1.6	1994 ± 68
	validation-selected	0.82 ± 0.17	20.3 ± 1.5	3.29 ± 0.16	32.0 ± 6.7	1230 ± 299
PPO-C · drawdown only (λ_1=0, λ_2=1)	50k	1.25 ± 0.14	18.9 ± 1.8	3.17 ± 0.15	29.9 ± 1.3	1111 ± 60
	100k	1.01 ± 0.23	21.5 ± 2.2	3.46 ± 0.34	45.6 ± 0.5	1938 ± 44
	150k (final)	0.95 ± 0.17	23.4 ± 4.5	3.68 ± 0.29	53.6 ± 0.3	2431 ± 55
	validation-selected	1.23 ± 0.13	17.7 ± 0.9	3.07 ± 0.10	20.1 ± 5.8	730 ± 259
PPO-D · CVaR + drawdown (λ_1=1, λ_2=1)	50k	1.24 ± 0.20	18.6 ± 0.9	3.09 ± 0.27	27.9 ± 0.5	1014 ± 29
	100k	1.09 ± 0.25	21.1 ± 0.5	3.21 ± 0.31	41.8 ± 1.0	1692 ± 76
	150k (final)	1.06 ± 0.22	22.7 ± 3.8	3.23 ± 0.06	46.8 ± 2.0	1959 ± 128
	validation-selected	1.25 ± 0.14	18.3 ± 1.6	3.07 ± 0.16	19.1 ± 2.2	654 ± 84
PPO-E · calibrated CVaR + DD (λ_1=0.05, λ_2=0.5)	50k	1.18 ± 0.08	19.9 ± 1.2	3.18 ± 0.06	30.8 ± 1.5	1155 ± 43
	100k	1.00 ± 0.14	22.6 ± 1.1	3.49 ± 0.13	48.9 ± 1.2	2134 ± 37
	150k (final)	0.87 ± 0.14	23.7 ± 2.6	3.77 ± 0.29	57.5 ± 1.4	2660 ± 11
	validation-selected	1.21 ± 0.10	18.4 ± 1.2	3.10 ± 0.07	19.8 ± 5.9	700 ± 246

Table 27: Simulated market descriptive statistics (3,000 days; 16.6% stress; 18 stress episodes, mean length 27.7 days). Asset labels are nominal. *Note:* Average pairwise correlation 0.27 in calm and 0.61 in stress regimes.

Asset	Ann. mean (%)	Ann. vol (%)	Vol, calm (%)	Vol, stress (%)	Skew	Excess kurt.	Worst day (%)
EQ_US	3.51	44.20	35.03	74.77	1.22	26.90	−22.85
EQ_EU	17.50	37.08	29.66	62.04	0.26	6.06	−16.04
EQ_EM	6.12	36.57	28.56	62.80	−0.33	4.20	−14.87
BOND_UST	−0.41	14.72	11.52	25.06	0.04	5.34	−5.73
COMMODITY	3.81	24.79	19.23	42.89	0.32	4.20	−9.66

Table 28: Environment, friction and reward parameters.

Parameter	Value	Parameter	Value
Initial NAV (per fold)	\$10,000,000	Look-back window L	20 days
Execution	decide at close t; share order filled at open t+1	Cash rate	0%
Commission c / half-spread s	1 bp / 2 bp	Temporary impact η	0.15
Permanent impact γ / half-life	0.05 / 5 days	Participation cap ρ	10% of ADV
ADV / volatility windows	20 / 20 days	Action map	softmax with cash logit, $\kappa = 3$
CVaR level α / window W	0.95 / 63 days (min. 20)	Reward scale k / clip	100 / ±10
Smoothed execution κ_{tr} / band b	0.1 / 1% (S, SE)	Penalties (E, SE)	$\lambda_1 = 0.05$, $\lambda_2 = 0.5$